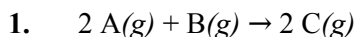


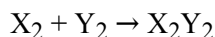
Chapter 09 Part 3: Reaction Mechanism



When the concentration of substance B in the reaction above is doubled, all other factors being held constant, it is found that the rate of the reaction remains unchanged. The most probable explanation for this observation is that

- (A) the order of the reaction with respect to substance B is 1
- (B) substance B is not involved in any of the steps in the mechanism of the reaction
- (C) substance B is not involved in the rate-determined step of the mechanism, but is involved in subsequent steps ✓
- (D) substance B is probably a catalyst, and as such, its effect on the rate of the reaction does not depend on its concentration
- (E) the reactant with the smallest coefficient in the balanced equation generally has little or no effect on the rate of the reaction

2.



$$\text{rate} = k[\text{X}_2]$$

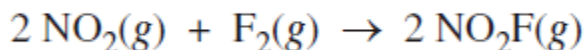
A reaction and its experimentally determined rate law are represented above. A chemist proposes two different possible mechanisms for the reaction, which are given below.

Mechanism 1	Mechanism 2
$\text{X}_2 \rightarrow 2 \text{X}$ (<i>slow</i>)	$\text{X}_2 \rightarrow 2 \text{X}$ (<i>slow</i>)
$\text{X} + \text{Y}_2 \rightarrow \text{XY}_2$ (<i>fast</i>)	$\text{X} + \text{Y}_2 \rightarrow \text{XY} + \text{Y}$ (<i>fast</i>)
$\text{X} + \text{XY}_2 \rightarrow \text{X}_2\text{Y}_2$ (<i>fast</i>)	$\text{X} + \text{XY} \rightarrow \text{X}_2\text{Y}$ (<i>fast</i>)
	$\text{X}_2\text{Y} + \text{Y} \rightarrow \text{X}_2\text{Y}_2$ (<i>fast</i>)

Based on the information above, which of the following is true?

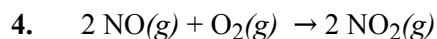
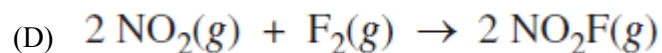
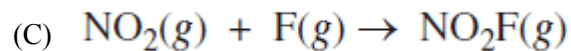
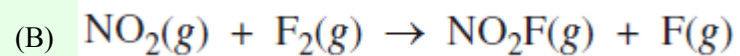
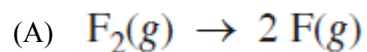
- (A) Only mechanism 1 is consistent with the rate law.
- (B) Only mechanism 2 is consistent with the rate law.
- (C) Both mechanism 1 and mechanism 2 are consistent with the rate law. ✓
- (D) Neither mechanism 1 nor mechanism 2 is consistent with the rate law.

3.

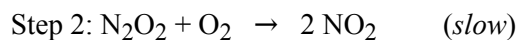
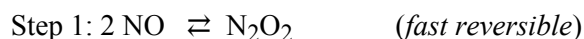


The rate law for the reaction represented by the equation above is $\text{rate} = k[\text{NO}_2][\text{F}_2]$. Which of the following could be the first elementary step of a two-step mechanism for the reaction if the first step is slow and the second step is fast?

Chapter 09 Part 3: Reaction Mechanism



Consider the following mechanism for the reaction represented above.



Which of the following statements is true?

(A) Step 1 represents a unimolecular reaction.

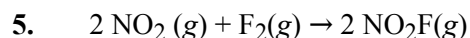
(B) Increasing the concentration of NO will decrease the overall rate of the reaction.

(C) Raising the temperature will have no effect on the numerical value of the rate constant.

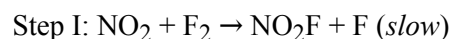
(D) The rate law that is consistent with the mechanism is $rate = k[\text{NO}]^2[\text{O}_2]$.



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It is proposed that the reaction represented above proceeds via the mechanism represented by the two elementary steps shown below.



- a. Step I of the proposed mechanism involves the collision between NO_2 and F_2 molecules. This step is slow even though such collisions occur very frequently in a mixture of $\text{NO}_2(g)$ and $\text{F}_2(g)$. Consider a specific collision between a molecule of NO_2 and a molecule of F_2 .
- One factor that affects whether the collision will result in a reaction is the magnitude of the collision energy. Explain.
 - Identify and explain one other factor that affects whether the collision will result in a reaction.
- b. Consider the following potential rate laws for the reaction. Circle the rate law below that is consistent with the mechanism proposed above. Explain the reasoning behind your choice in terms of the details of the elementary steps of the mechanism.

$$\text{rate} = k[\text{NO}_2]^2[\text{F}_2] \quad \text{rate} = k[\text{NO}_2][\text{F}_2]$$

Part A

1 point is earned for a correct explanation that makes reference to the activation energy of the reaction.

Successful molecular collisions must have sufficient energy in order to result in reaction. Only collisions with sufficient energy to overcome the activation energy barrier, E_a , will be able to reach the transition state and begin to break the F–F bond.

1 point is earned for identifying the relative orientation of the colliding molecules.

For a collision to be successful, the molecules must have the correct orientation.

1 point is earned for an explanation that makes reference to specific parts (atoms or bonds) of the reacting molecules.

Only collisions with the correct orientation will be able to begin to form an N–F bond and begin to break an F–F bond as the transition state is approached (that is, the molecules must contact each other at very specific locations on their surfaces for the transition state to be accessible).



0	1	2	3
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The student response earns three of the following points:

Chapter 09 Part 3: Reaction Mechanism

1 point is earned for a correct explanation that makes reference to the activation energy of the reaction.

Successful molecular collisions must have sufficient energy in order to result in reaction. Only collisions with sufficient energy to overcome the activation energy barrier, E_a , will be able to reach the transition state and begin to break the F–F bond.

1 point is earned for identifying the relative orientation of the colliding molecules.

For a collision to be successful, the molecules must have the correct orientation.

1 point is earned for an explanation that makes reference to specific parts (atoms or bonds) of the reacting molecules.

Only collisions with the correct orientation will be able to begin to form an N–F bond and begin to break an F–F bond as the transition state is approached (that is, the molecules must contact each other at very specific locations on their surfaces for the transition state to be accessible).

Part B

1 point is earned for identifying the correct rate law with a proper explanation.

(The explanation must correlate the overall rate law with the rate law derived from the stoichiometry of the slow step in the mechanism).

Note: A statement relating the coefficients of the reactants in Step I to the exponents in the rate law indicates a correct understanding.

The rate law that is consistent with the mechanism is the one on the right above ($\text{rate} = k[\text{NO}_2][\text{F}_2]$).

Step I is the slower step and the rate-determining step in the mechanism. Since Step I is an elementary reaction, its rate law is given by the stoichiometry of the reacting molecules,

$$\text{rate}_{\text{Step I}} = k[\text{NO}_2][\text{F}_2].$$



0	1
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The student response earns one of the following points:

1 point is earned for identifying the correct rate law with a proper explanation.

(The explanation must correlate the overall rate law with the rate law derived from the stoichiometry of the slow step in the mechanism).

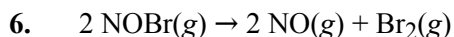
Note: A statement relating the coefficients of the reactants in Step I to the exponents in the rate law indicates a correct understanding.

The rate law that is consistent with the mechanism is the one on the right above ($\text{rate} = k[\text{NO}_2][\text{F}_2]$).

Step I is the slower step and the rate-determining step in the mechanism. Since Step I is an elementary reaction, its rate law is given by the stoichiometry of the reacting molecules,

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$$\text{rate}_{\text{Step I}} = k[\text{NO}_2][\text{F}_2].$$



The equation above represents an elementary step in a chemical reaction. Which of the following is the correct expression for the rate law of the elementary step?

(A) $\text{Rate} = k[\text{NOBr}]^{1/2}$

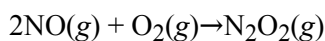
(B) $\text{Rate} = k[\text{NOBr}]$

(C) $\text{Rate} = k[\text{NOBr}]^2$

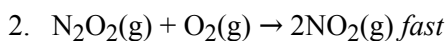
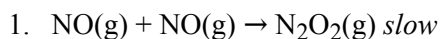
(D) $\text{Rate} = k[\text{NO}]^2 [\text{Br}_2]$



7.



A possible mechanism for the overall reaction represented above is the following.



Which of the following rate expressions agrees best with this possible mechanism?

(A) $\text{Rate} = k[\text{NO}]^2$

(B) $\text{Rate} = k \frac{[\text{NO}]}{[\text{O}_2]}$

(C) $\text{Rate} = k \frac{[\text{NO}]^2}{[\text{O}_2]}$

(D) $\text{Rate} = k[\text{NO}]^2 [\text{O}_2]$

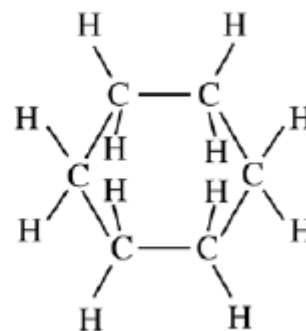
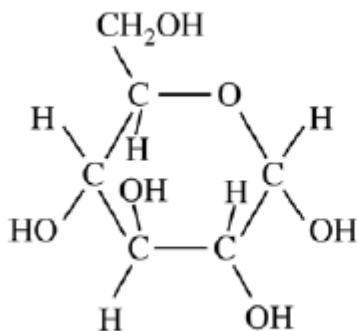
(E) $\text{Rate} = k[\text{N}_2\text{O}_2] [\text{O}_2]$



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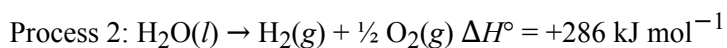
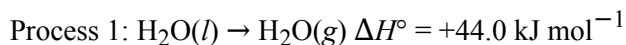
8. Answer each of the following in terms of principles of molecular behavior and chemical concepts.

- a. The structures for glucose, $C_6H_{12}O_6$, and cyclohexane, C_6H_{12} , are shown below.



Identify the type(s) of intermolecular attractive forces in

- pure glucose
 - pure cyclohexane
- b. Glucose is soluble in water but cyclohexane is not soluble in water. Explain.
- c. Consider the two processes represented below.

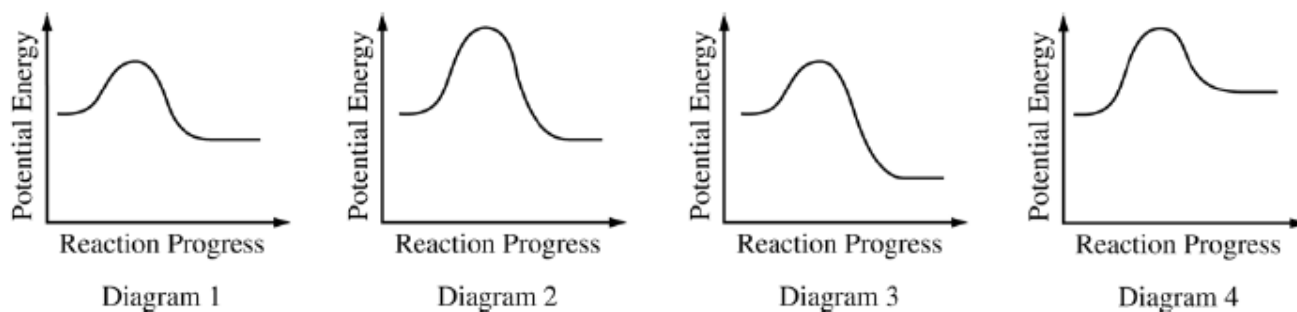


- For each of the two processes, identify the type(s) of intermolecular or intramolecular attractive forces that must be overcome for the process to occur.
- Indicate whether you agree or disagree with the statement in the box below. Support your answer with a short explanation.

When water boils, H_2O molecules break apart to form hydrogen molecules and oxygen molecules.

- d. Consider the four reaction-energy profile diagrams shown below.

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- Identify the two diagrams that could represent a catalyzed and an uncatalyzed reaction pathway for the same reaction. Indicate which of the two diagrams represents the catalyzed reaction pathway for the reaction.
- Indicate whether you agree or disagree with the statement in the box below. Support your answer with a short explanation.

Adding a catalyst to a reaction mixture adds energy that causes the reaction to proceed more quickly.

Part A

- 1 point is earned for the correct answer.

Hydrogen bonding OR dipole-dipole interactions OR van der Waals interactions (London dispersion forces may also be mentioned.)

- 1 point is earned for London dispersion forces.

London dispersion forces



0	1	2
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The student response earns two of the following points:

- 1 point is earned for the correct answer.

Hydrogen bonding OR dipole-dipole interactions OR van der Waals interactions (London dispersion forces may also be mentioned.)

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- 1 point is earned for London dispersion forces.

London dispersion forces

Part B

1 point is earned for explaining the solubility of glucose in terms of hydrogen bonding or dipole-dipole interactions with water.

1 point is earned for explaining the difference in the polarity of cyclohexane and water.

The hydroxyl groups in glucose molecules can form strong hydrogen bonds with the solvent (water) molecules, so glucose is soluble in water. In contrast, cyclohexane is not capable of forming strong intermolecular attractions with water (no hydrogen bonding), so the water-cyclohexane interactions are not as energetically favorable as the interactions that already exist among polar water molecules.

OR

1 point is earned for any one of the three concepts; two points are earned for any two of the three concepts.

- Glucose is polar and cyclohexane is nonpolar.
- Polar solutes (such as glucose) are generally soluble in polar solvents such as water.
- Nonpolar solutes (such as cyclohexane) are not soluble in the polar solvent.



0	1	2
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The student response earns two of the following points:

1 point is earned for explaining the solubility of glucose in terms of hydrogen bonding or dipole-dipole interactions with water.

1 point is earned for explaining the difference in the polarity of cyclohexane and water.

The hydroxyl groups in glucose molecules can form strong hydrogen bonds with the solvent (water) molecules, so glucose is soluble in water. In contrast, cyclohexane is not capable of forming strong intermolecular attractions with water (no hydrogen bonding), so the water-cyclohexane interactions are not as energetically favorable as the interactions that already exist among polar water molecules.

OR

1 point is earned for any one of the three concepts; two points are earned for any two of the three concepts.

- Glucose is polar and cyclohexane is nonpolar.
- Polar solutes (such as glucose) are generally soluble in polar solvents such as water.
- Nonpolar solutes (such as cyclohexane) are not soluble in the polar solvent.

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Part C

- 1 point is earned for identifying the type of intermolecular force involved in process 1.

In process 1, hydrogen bonds (or dipole-dipole interactions) in liquid water are overcome to produce distinct water molecules in the vapor phase.

1 point is earned for identifying the type of intramolecular bonding involved in process 2.

In process 2, covalent bonds (or sigma bonds, or electron-pair bonds) within water molecules must be broken to allow the atoms to recombine into molecular hydrogen and oxygen.

- 1 point is earned for disagreeing with the statement and providing a correct explanation.

I disagree with the statement. Boiling is simply Process 1, in which only intermolecular forces are broken and the water molecules stay intact. No intramolecular or covalent bonds break in this process.



0	1	2	3
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The student response earns three of the following points:

- 1 point is earned for identifying the type of intermolecular force involved in process 1.

In process 1, hydrogen bonds (or dipole-dipole interactions) in liquid water are overcome to produce distinct water molecules in the vapor phase.

1 point is earned for identifying the type of intramolecular bonding involved in process 2.

In process 2, covalent bonds (or sigma bonds, or electron-pair bonds) within water molecules must be broken to allow the atoms to recombine into molecular hydrogen and oxygen.

- 1 point is earned for disagreeing with the statement and providing a correct explanation.

I disagree with the statement. Boiling is simply Process 1, in which only intermolecular forces are broken and the water molecules stay intact. No intramolecular or covalent bonds break in this process.

Part D

- 1 point is earned for identifying the correct graphs and indicating which represents which pathway.

Diagram 1 represents a catalyzed pathway and diagram 2 represents an uncatalyzed pathway for the same reaction.

- 1 point is earned for disagreeing with the statement and providing an explanation.

I disagree with the statement. A catalyst does not add energy, but provides an alternate reaction pathway with a lower activation energy.

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0	1	2
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The student response earns two of the following points:

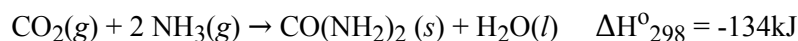
- 1 point is earned for identifying the correct graphs and indicating which represents which pathway.

Diagram 1 represents a catalyzed pathway and diagram 2 represents an uncatalyzed pathway for the same reaction.

- 1 point is earned for disagreeing with the statement and providing an explanation.

I disagree with the statement. A catalyst does not add energy, but provides an alternate reaction pathway with a lower activation energy.

9. Answer the following questions in terms of thermodynamic principles and concepts of kinetic molecular theory.
- (a) Consider the reaction represented below, which is spontaneous at 298 K.



- (i) For the reaction, indicate whether the standard entropy change, ΔS°_{298} is positive, or negative, or zero. Justify your answer.
- (ii) Which factor, the change in enthalpy, ΔH°_{298} or the change in entropy, ΔS°_{298} , provides the principal driving force for the reaction at 298 K? Explain.
- (iii) For the reaction, how is the value of the standard free energy change, ΔG° , affected by an increase in temperature? Explain.
- (b) Some reactions that are predicted by their sign of ΔG° to be spontaneous at room temperature do not proceed at a measurable rate at room temperature.
- (i) Account for this apparent contradiction.
- (ii) A suitable catalyst increases the rate of such a reaction. What effect does the catalyst have on ΔG° for the reaction? Explain.

Part A (i)

1 point earned for correct identification of (-) sign of ΔS°

1 point earned for correct explanation (mention of phases is crucial for point)

No point earned if incorrect ΔS° sign is obtained from the presumed value of ΔG°

ΔS° is negative (-) OR $\Delta S^\circ < 0$ OR entropy is decreasing.

3 moles of gaseous particles are converted to 2 moles of solid/liquid

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0	1	2
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The student response earns two of the following points:

1 point earned for correct identification of (-) sign of ΔS_o

1 point earned for correct explanation (mention of phases is crucial for point)

No point earned if incorrect ΔS_o sign is obtained from the presumed value of ΔG_o

ΔS_o is negative (-) OR $\Delta S_o < 0$ OR entropy is decreasing.

3 moles of gaseous particles are converted to 2 moles of solid/liquid

Part A (ii)

1 point is earned for identifying ΔH_o as the principal driving force for the reaction

1 point earned for correct justification

ΔH_o drives the reaction.

The decrease in entropy ($\Delta S_o < 0$) cannot drive the reaction, so the decrease in enthalpy

($\Delta H_o < 0$) MUST drive the reaction.

OR

$\Delta G_o = \Delta H_o - T\Delta S_o$; for a spontaneous reaction $\Delta G_o < 0$, and a negative value of ΔS_o causes a positive ΔG_o .



0	1	2
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The student response earns two of the following points:

1 point is earned for identifying ΔH_o as the principal driving force for the reaction

1 point earned for correct justification

ΔH_o drives the reaction.

The decrease in entropy ($\Delta S_o < 0$) cannot drive the reaction, so the decrease in enthalpy

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($\Delta H_o < 0$) MUST drive the reaction.

OR

$\Delta G_o = \Delta H_o - T\Delta S_o$; for a spontaneous reaction $\Delta G_o < 0$, and a negative value of ΔS_o causes a positive ΔG_o .

A (iii)

1 point earned for the description of the effect of an increase in temperature on ΔS_o and consequently on ΔG_o

No point earned for an argument based on Le Chatelier's principle

Given that $\Delta G_o = \Delta H_o - T\Delta S_o$ and $\Delta S_o < 0$, an increase in temperature causes an increase in the value of ΔG_o (ΔG_o becomes less negative).



0	1
---	---

The student response earns one of the following points:

1 point earned for the description of the effect of an increase in temperature on ΔS_o and consequently on ΔG_o

No point earned for an argument based on Le Chatelier's principle

Given that $\Delta G_o = \Delta H_o - T\Delta S_o$ and $\Delta S_o < 0$, an increase in temperature causes an increase in the value of ΔG_o (ΔG_o becomes less negative).

Part B (i)

1 point is earned for linking the rate of the reaction to the activation energy, which may be explained verbally or shown on a reaction profile diagram

The reaction rate depends on the reaction kinetics, which is determined by the value of the activation energy, E_{act} . If the activation energy is large, a reaction that is thermodynamically spontaneous may proceed very slowly (if at all).

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0	1
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The student response earns one of the following points:

1 point is earned for linking the rate of the reaction to the activation energy, which may be explained verbally or shown on a reaction profile diagram

The reaction rate depends on the reaction kinetics, which is determined by the value of the activation energy, E_{act} . If the activation energy is large, a reaction that is thermodynamically spontaneous may proceed very slowly (if at all).

Part B (ii)

=1 point earned for indicating no change in the value of ΔG_o

1 point earned for indicating (verbally, or with a reaction-profile diagram) that the catalyst affects the activation energy

The catalyst has no effect on the value of ΔG_o .

The catalyst reduces the value of E_{act} , increasing the rate of reaction, but has no effect on the values of ΔH_o and ΔS_o , so it cannot affect the thermodynamics of the reaction.



0	1	2
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The student response earns two of the following points:

1 point earned for indicating no change in the value of ΔG_o

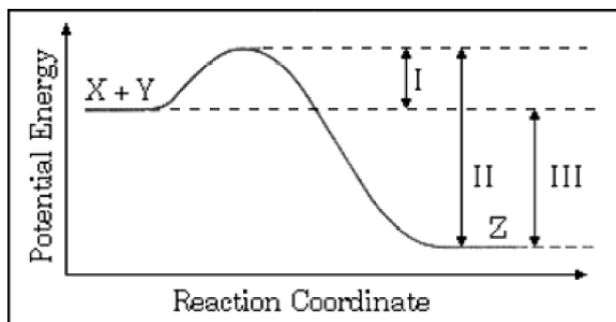
1 point earned for indicating (verbally, or with a reaction-profile diagram) that the catalyst affects the activation energy

The catalyst has no effect on the value of ΔG_o .

The catalyst reduces the value of E_{act} , increasing the rate of reaction, but has no effect on the values of ΔH_o and ΔS_o , so it cannot affect the thermodynamics of the reaction.

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10.



The energy diagram for the reaction $X + Y \rightarrow Z$ is shown above. The addition of a catalyst to this reaction would cause a change in which of the indicated energy differences?

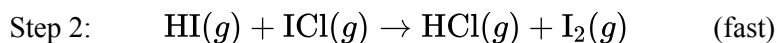
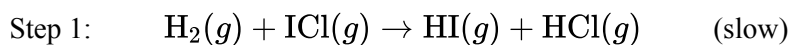
- (A) I only
(B) II only
(C) III only
(D) I and II only
(E) I, II, and III



11. Step 1: $O_3(g) + Cl(g) \rightarrow O_2(g) + ClO(g)$ Step 2: $ClO(g) + O(g) \rightarrow O_2(g) + Cl(g)$

A proposed mechanism for the destruction of $O_3(g)$ in the upper atmosphere is represented above. Which of the following is the catalyst in the proposed mechanism?

- (A) $O(g)$
(B) $Cl(g)$
(C) $ClO(g)$
(D) $O_2(g)$



12. Which of the following represents the overall chemical equation for the reaction and the rate law for elementary step 2?

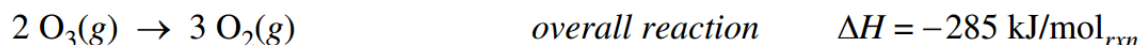
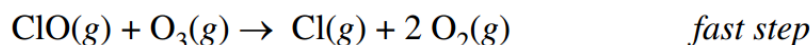
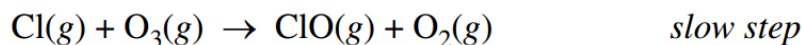
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- (A) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{HI}][\text{ICl}]$.
- (B) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{ICl}]^2$.
- (C) The overall reaction is $\text{H}_2(g) + \text{ICl}(g) \rightarrow \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{H}_2][\text{ICl}]$.
- (D) The overall reaction is $\text{H}_2(g) + 2 \text{ICl}(g) \rightarrow 2 \text{HCl}(g) + \text{I}_2(g)$; the rate law for step 2 is $\text{rate} = k[\text{HI}][\text{ICl}]$. ✓

13. Which of the following represents a rate law for the overall reaction that is consistent with the proposed mechanism?

- (A) $\text{Rate} = k[\text{H}_2][\text{ICl}]$ ✓
- (B) $\text{Rate} = k[\text{H}_2][\text{ICl}]^2$
- (C) $\text{Rate} = k[\text{H}_2][\text{HI}][\text{ICl}]^2$
- (D) $\text{Rate} = k[\text{HI}][\text{ICl}]$

When free $\text{Cl}(g)$ atoms encounter $\text{O}_3(g)$ molecules in the upper atmosphere, the following reaction mechanism is proposed to occur.

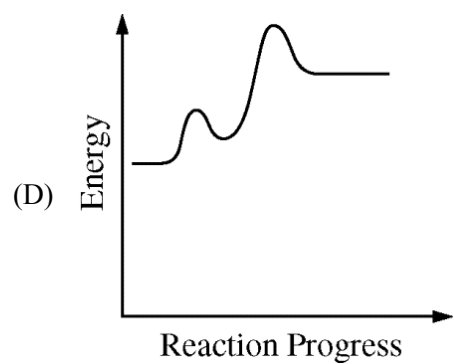
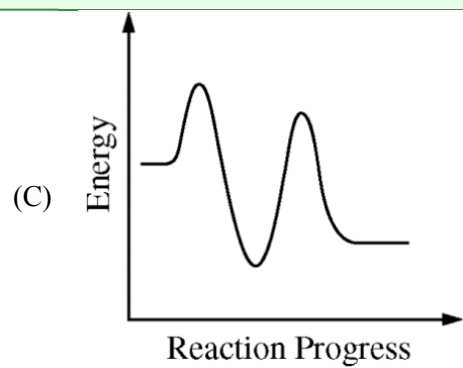
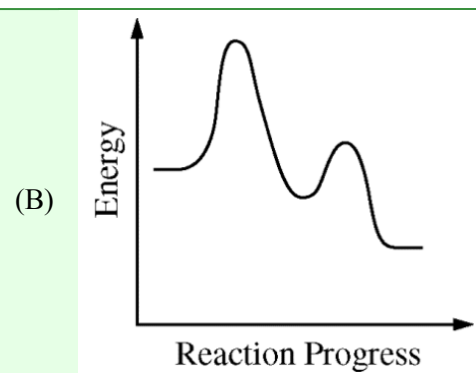
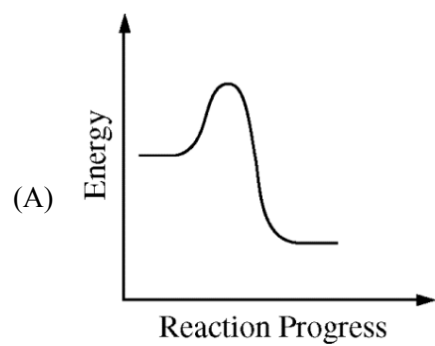


14. Which of the following rate laws for the overall reaction corresponds to the proposed mechanism?

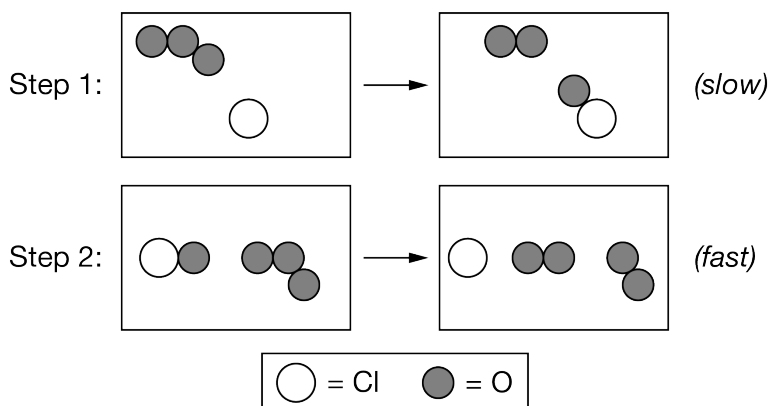
- (A) $\text{Rate} = k[\text{O}_3]^2$
- (B) $\text{Rate} = k[\text{Cl}][\text{O}_3]$ ✓
- (C) $\text{Rate} = k[\text{ClO}][\text{O}_3]^2$
- (D) $\text{Rate} = k[\text{O}_2]^3 / [\text{O}_3]^2$

15. Which of the following reaction energy profiles best corresponds to the proposed mechanism?

Chapter 09 Part 3: Reaction Mechanism



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The particle models shown above represent a proposed two-step mechanism for the destruction of ozone (O_3) in the upper atmosphere.

16.

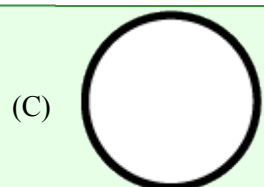
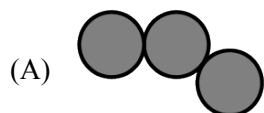


Based on the proposed mechanism, which of the following best describes the concentration of the species represented above as the reaction occurs?

- (A) The concentration is larger than that of reactant or product.
- (B) The concentration is very low for the duration of the reaction. ✓
- (C) The concentration increases steadily as the reaction proceeds.
- (D) The concentration fluctuates between high and low during the reaction.

17. Based on the models, which of the following represents a species that acts as a catalyst for the reaction?

Chapter 09 Part 3: Reaction Mechanism



19. Based on the proposed mechanism, which of the following is the rate-law expression for the destruction of O_3 ?

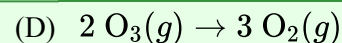
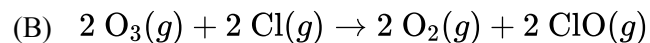
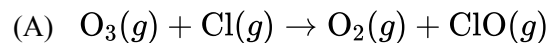
(A) $\text{Rate} = k[\text{O}_3]^2$

(B) $\text{Rate} = k[\text{Cl}]^2$

(C) $\text{Rate} = k[\text{ClO}][\text{O}_3]$

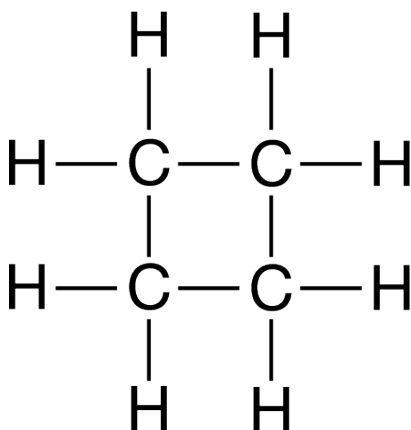
(D) $\text{Rate} = k[\text{O}_3][\text{Cl}]$

18. Based on the proposed mechanism, what is the balanced chemical equation for the overall reaction?



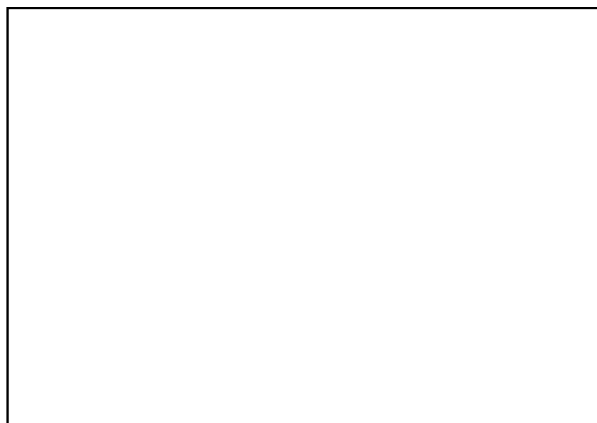
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20. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



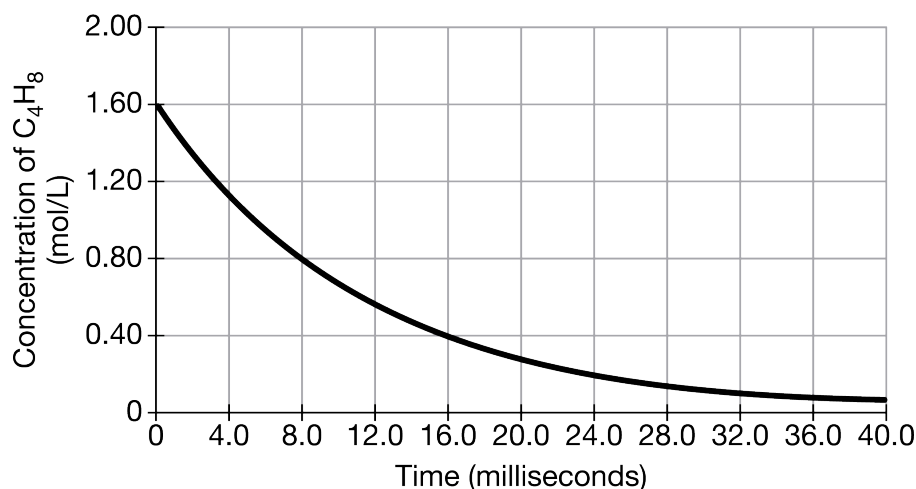
The hydrocarbon cyclobutane, C_4H_8 , is represented above. At high temperatures, cyclobutane quickly decomposes into ethene, C_2H_4 .

- (a) Draw a Lewis electron-dot diagram of the ethene molecule in the following box.



- (b) Estimate the value of the $\text{H} - \text{C} - \text{H}$ bond angle in ethene.
- (c) The graph below shows the concentration of cyclobutane as it decomposes over time at 1270 K.

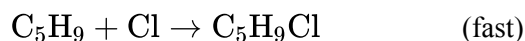
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(i) Based on the graph, determine the order of the decomposition reaction of cyclobutane at 1270 K. Justify your answer.

(ii) Calculate the time, in milliseconds, that it would take for 99 percent of the original cyclobutane at 1270 K to decompose.

Another alkane with a ring structure is cyclopentane, C₅H₁₀(g). Cyclopentane reacts with chlorine, Cl₂(g), to produce C₅H₉Cl(g) and HCl(g). Following is a representation of a proposed mechanism for the reaction.



(d) Write a rate law for the reaction that is consistent with the mechanism. Justify your answer.

(e) A student claims that Cl₂(g) is a catalyst in the reaction. Explain why the student's claim is false.

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The diagram meets all of the following criteria:

- ☐ There are two carbon atoms connected by a double bond.
- ☐ Each carbon atom has two hydrogen atoms connected to it by single bonds.

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- ☐ There are no nonbonding electrons on any of the atoms.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The actual bond angle is 117° .



0	1
---	---

The response indicates that the bond angle is $120^\circ (\pm 5^\circ)$ OR about 120° .

Part (c)(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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The response indicates both of the following criteria:

- ☐ The response indicates that the reaction is first order
- ☐ The response gives a correct justification (e.g., the half-life is constant OR the concentration is halved in 8.0 milliseconds and halved again in another 8.0 milliseconds).

Part (c)(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response. The student earns the second point if an incorrect rate constant is used correctly to calculate the time.



0	1	2
---	---	---

The response gives both of the following calculations (or variations). (One point is earned for the calculation of the half-life and the second point for the calculation of the time):

- ☐ $t_{1/2} = \frac{0.693}{k}$
 $k = \frac{0.693}{t_{1/2}} = \frac{0.693}{8.0 \text{ milliseconds}} = 0.087 \text{ millisecond}^{-1}$
- ☐ $\ln[A]_t - \ln[A]_0 = -kt$
 $\ln(1.60 \times 0.01) - \ln(1.60) = -kt$
 $t = \frac{\ln(0.016) - \ln(1.60)}{-0.087 \text{ millisecond}^{-1}} = 53 \text{ milliseconds}$

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Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
---	---	---

The response indicates both of the following criteria:

- ☐ The response indicates that the rate law is $\text{rate} = k[\text{Cl}_2]$
- ☐ The response justifies the answer by indicating that the first step is the rate-determining (or slow) step and the reactant in that step is Cl_2 .

Part (e)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

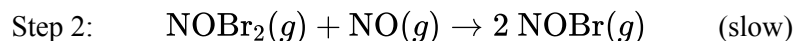


0	1
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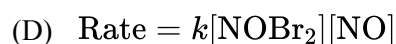
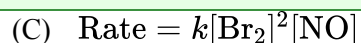
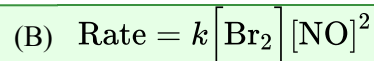
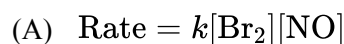
The response indicates that a catalyst is not consumed in a reaction but Cl_2 is consumed in this reaction.



The reaction represented by the equation above has the following proposed mechanism.

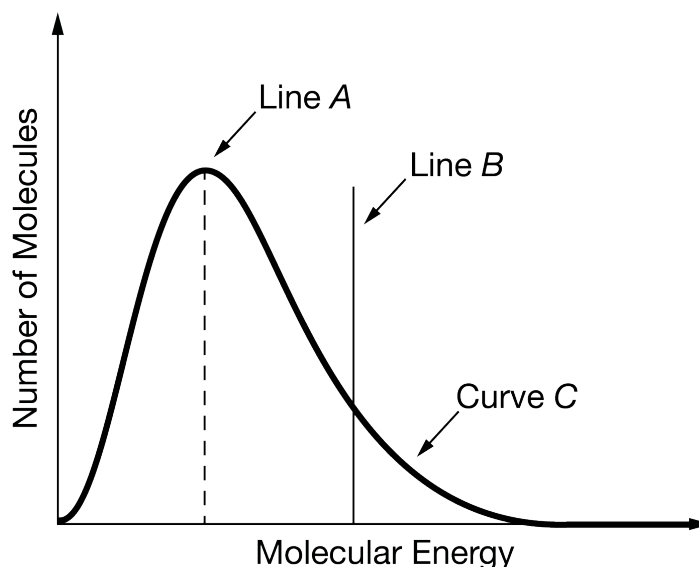


Based on the information, which of the following is the initial rate law for the reaction?



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22.



The energy distribution profile (Curve C) for the Y_2 molecules is shown in the graph above for the reaction $\text{X} + \text{Y}_2 \rightarrow \text{XY}_2$ when it is done under certain experimental conditions. Line A represents the most probable energy of the Y_2 molecules, and Line B represents the activation energy. Which of the following changes should be made to the graph to explain the increase in the rate of the reaction if the only change in experimental conditions is the addition of a catalyst?

- (A) Curve C will broaden and Line B will move to the left, because more Y_2 molecules will have an energy greater than the minimum energy needed to overcome the activation energy barrier.
- (B) Line B will move to the left because a larger fraction of the Y_2 molecules will have the minimum energy to overcome the activation energy barrier. ✓
- (C) Curve C and Line A will move to the right because the average energy of the Y_2 molecules will increase.
- (D) Line A will move to the right because the most probable energy of the Y_2 molecules will increase.

23. To catalyze a biochemical reaction, an enzyme typically

- (A) drives the reaction to completion by consuming byproducts of the reaction
- (B) binds temporarily to reactant molecules to lower the activation energy of the reaction ✓
- (C) dissociates into additional reactant molecules, thereby increasing the reaction rate
- (D) decomposes and releases energy to increase the number of successful collisions between reactant molecules

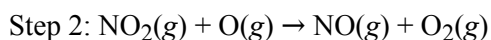
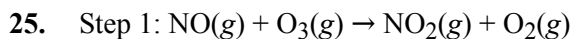
24. Step 1: $\text{Cl}(g) + \text{O}_3(g) \rightarrow \text{ClO}(g) + \text{O}_2(g)$

Step 2: $\text{ClO}(g) + \text{O}(g) \rightarrow \text{Cl}(g) + \text{O}_2(g)$

A proposed mechanism for destruction of ozone gas in the stratosphere is represented above. Which of the following is evidence that the mechanism is occurring?

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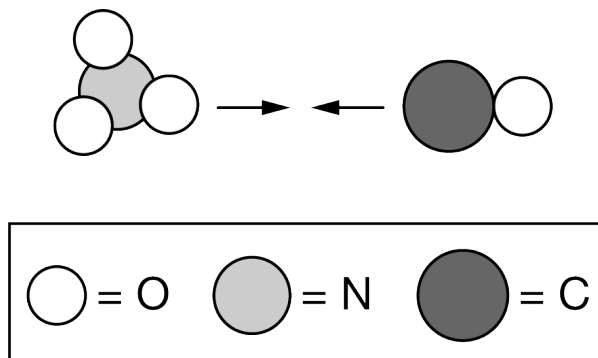
- (A) The presence of Cl(g) increases the rate of the overall reaction. ✓
- (B) The presence of Cl(g) decreases the rate of the overall reaction.
- (C) The presence of Cl(g) increases the equilibrium constant of the overall reaction.
- (D) The presence of Cl(g) decreases the equilibrium constant of the overall reaction.



A reaction mechanism for the destruction of ozone, $\text{O}_3\text{(g)}$, is represented above. In the overall reaction, NO(g) is best described as

- (A) an inhibitor
- (B) a catalyst ✓
- (C) a reactant
- (D) an intermediate
- (E) a product

26.



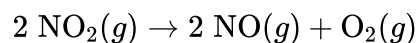
The slowest step in a reaction mechanism requires the collision represented above to occur. Which of the following most likely indicates how the addition of a solid catalyst could increase the rate of the reaction?

- (A) The catalyst could change the reaction from second order to third order.
- (B) The catalyst could increase the particles' speed, thereby increasing the value of the rate constant, k .
- (C) The catalyst could decrease the particles' speed, making it less likely that the particles will rebound without reacting when they collide.
- (D) The catalyst could adsorb one of the particles, making a successful (reaction-producing) collision with the other particle more likely. ✓

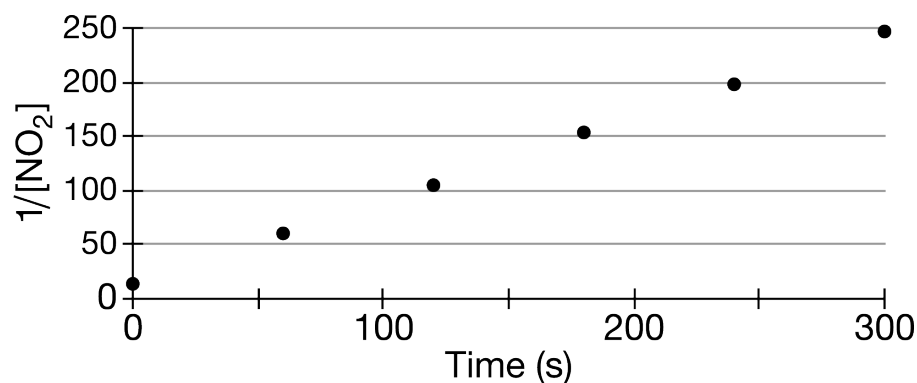
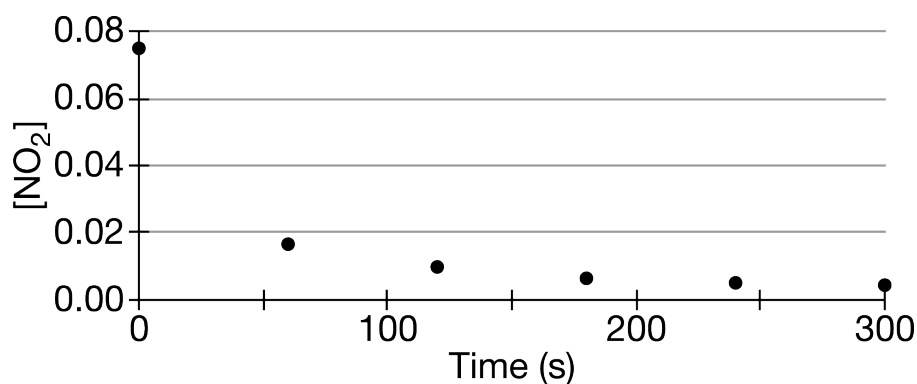
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27. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

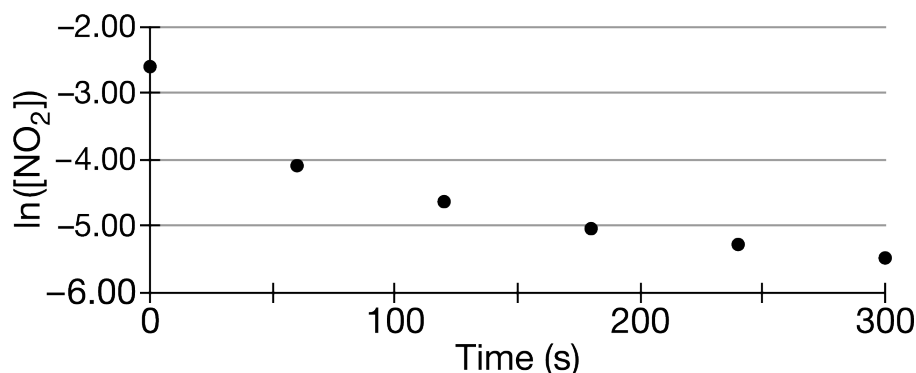
Nitrogen dioxide, $\text{NO}_2(g)$, is produced as a by-product of the combustion of fossil fuels in internal combustion engines. At elevated temperatures $\text{NO}_2(g)$ decomposes according to the equation below.



The concentration of a sample of $\text{NO}_2(g)$ is monitored as it decomposes and is recorded on the graph directly below. The two graphs that follow it are derived from the original data.

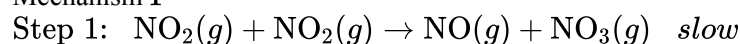


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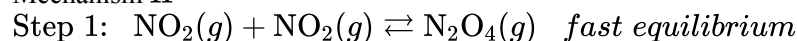
- (a) Explain how the graphs indicate that the reaction is second order.
- (b) Write the rate law for the decomposition of $\text{NO}_2(g)$.
- (c) Consider two possible mechanisms for the decomposition reaction.
- (i) Is the rate law described by mechanism I shown below consistent with the rate law you wrote in part (b) ? Justify your answer.

Mechanism I



- (ii) Is the rate law described by mechanism II shown below consistent with the rate law you wrote in part (b) ? Justify your answer.

Mechanism II



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response indicates that the linear graph of $\frac{1}{[\text{NO}_2]}$ versus time indicates a second-order reaction.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response gives the following rate law.

$$\text{rate} = k[\text{NO}_2]^2$$

Part C(i)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ · The response indicates “Yes”.
- ☐ · The response gives a justification such as the following.
 Step 1 is slow; therefore, it is the rate-determining step of this mechanism.
 The rate law of this elementary reaction is $\text{rate} = k[\text{NO}_2][\text{NO}_2] = k[\text{NO}_2]^2$, which is consistent with the second-order rate law in part (b).

Part C(ii)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets both of the following criteria.

- ☐ · The response indicates “Yes”.
- ☐ · The response gives a justification such as the following.
 Step 2 is slow; therefore, it is the rate-determining step of this mechanism.
 The rate law of this elementary reaction is $\text{rate} = k[\text{N}_2\text{O}_4]$. Because N_2O_4 is an intermediate, it cannot appear in the rate law of the overall reaction. Because $K_{eq} = \frac{[\text{N}_2\text{O}_4]}{[\text{NO}_2]^2}$ in step 1,
 $[\text{N}_2\text{O}_4] = K_{eq}[\text{NO}_2]^2$. Then, substituting $K_{eq}[\text{NO}_2]^2$ for $[\text{N}_2\text{O}_4]$ in the rate law of step 2 gives
 $\text{rate} = (k K_{eq})[\text{NO}_2]^2$, which is consistent with the rate law in part (b).

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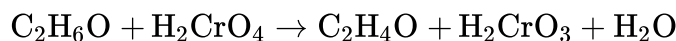
28. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, L and X, are shown in the following table.

Compound	L	X
Diagram		
Boiling Point	371 K	437 K

- (a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.
- (b) The C — O bond in compound L is longer than any C — H bond in compound X. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound L –
Compound X –
- (d) Explain why the boiling point of compound X is higher than that of compound L based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A 0.10 mol sample of compound L is placed in an evacuated, rigid container at 400. K and 0.25 atm. When more moles of compound L are added, the pressure increases to 0.75 atm at 400. K. Determine the number of moles of compound L now present in the container.

Another carbon-containing compound is $\text{C}_2\text{H}_6\text{O}$, which reacts with H_2CrO_4 , as represented by the following equation.



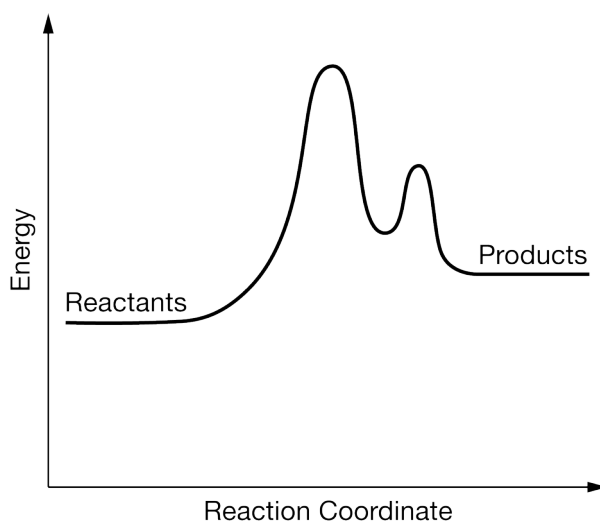
- (f) The student claims that C is oxidized when $\text{C}_2\text{H}_6\text{O}$ is converted to $\text{C}_2\text{H}_4\text{O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

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The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

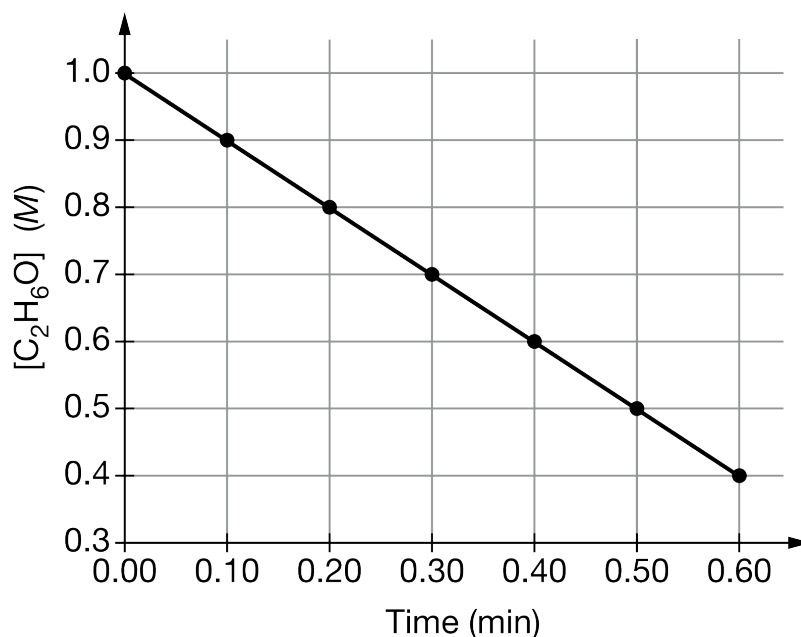
Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 1.0 M/min . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

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Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct hybridization.

· sp^3

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides a valid explanation equivalent to the following.

· The atomic radius of oxygen is larger than that of hydrogen because the valence electrons of oxygen occupy a higher shell than those of hydrogen ($n = 2$ versus $n = 1$). Thus, the internuclear distance (bond length) is longer for the C – O bond than for the C – H bond.

Part C

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response correctly identifies the intermolecular forces in both compounds.

- Compound L – dispersion forces, dipole-dipole forces, and hydrogen bonding
- Compound X – dispersion forces, dipole-dipole forces, and hydrogen bonding

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct explanation equivalent to the following.

- Although both compounds have the same types of intermolecular forces, the boiling point data indicate that the total of all intermolecular forces is greater in compound X than in compound L. The London dispersion forces in compound L and compound X are comparable due to their similar number of atoms, but compound X has a greater number of hydrogen bonding sites than compound L. Therefore, compound X has the higher boiling point.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct calculation.

$$\frac{n_1}{P_1} = \frac{n_2}{P_2}$$

$$\frac{0.10 \text{ mol}}{0.25 \text{ atm}} = \frac{n_2}{0.75 \text{ atm}}$$

$$n_2 = 0.30 \text{ mol}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- Agree. The oxidation number of carbon changes from -2 to -1 , indicating that it is oxidized.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

- No. If the second step of the mechanism is rate-determining, the second peak in the diagram should show a higher activation energy, not a lower activation energy, than the first step.

OR

- No. The higher energy of the products relative to the reactants is consistent with an endothermic reaction, not an exothermic reaction.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- Yes. Only a zeroth-order reaction will give a linear plot of $[\text{C}_2\text{H}_6\text{O}]$ versus time with a negative slope. The slope of the line is -1.0 M/min , which corresponds to a rate constant of 1.0 M/min since $\text{slope} = -k$ for a zeroth-order process.

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29. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, **L** and **Z**, are shown in the following table.

Compound	L	Z
Diagram	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{O}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{S}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $
Boiling Point	391 K	371 K

- (a) Based on VSEPR theory, what is the approximate value of the C — O — H bond angle in compound **L**?
- (b) The C — O bond in compound **L** is shorter than the C — S bond in compound **Z**. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.

Compound **L** –
 Compound **Z** –

- (d) Explain why the boiling point of compound **L** is higher than that of compound **Z** based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound **Z** is placed in an evacuated, rigid container at a temperature of 400. K and a pressure of 0.50 atm. Determine the final pressure when the temperature of the sample is increased to 600. K.

Another carbon-containing compound is $\text{C}_2\text{H}_6\text{O}$, which reacts with H_2CrO_4 , as represented by the following equation.



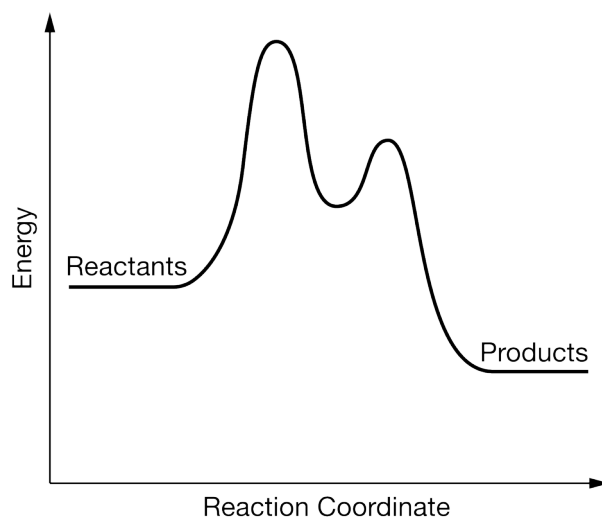
- (f) The student claims that **O** is oxidized when $\text{C}_2\text{H}_6\text{O}$ is converted to $\text{C}_2\text{H}_4\text{O}$. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

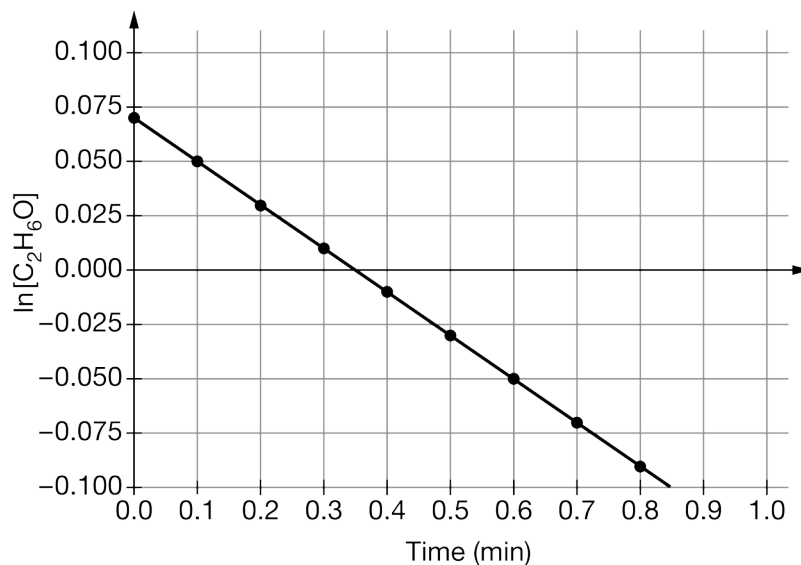
Chapter 09 Part 3: Reaction Mechanism

Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is first-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 0.50 min^{-1} . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides an acceptable bond angle.

- Any value between 104° and 110°

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

- The atomic radius of oxygen is smaller than that of sulfur because the valence electrons of oxygen occupy a lower shell than those of sulfur ($n = 2$ versus $n = 3$). Thus, the internuclear distance (bond length) is shorter for the C – O bond than for the C – S bond.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response correctly identifies the intermolecular forces in both compounds.

- Compound L – dispersion forces, dipole-dipole forces, and hydrogen bonding
- Compound Z – dispersion forces and dipole-dipole forces

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 09 Part 3: Reaction Mechanism

The response provides a correct explanation equivalent to the following.

· The boiling point of compound L is higher than that of compound Z because the intermolecular attractions in compound L are greater. The strong intermolecular hydrogen bonding in compound L, which is not present in compound Z, account for the overall stronger intermolecular attractions in compound L. Stronger attractions require more energy to be overcome, so compound L has the higher boiling point.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct calculation.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{0.50 \text{ atm}}{400. \text{ K}} = \frac{P_2}{600. \text{ K}}$$

$$P_2 = 0.75 \text{ atm}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The oxidation number of the oxygen atom remains unchanged at -2 , so it is neither oxidized nor reduced.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides correct answer and a valid justification equivalent to the following.

· No. The lower energy of the products relative to the reactants is consistent with an exothermic reaction, but the relative

Chapter 09 Part 3: Reaction Mechanism

height of the second peak is incorrect. If the second step of the mechanism is rate-determining, it should have an activation energy that is higher than that of the first step.

Part H

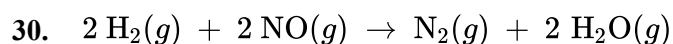
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



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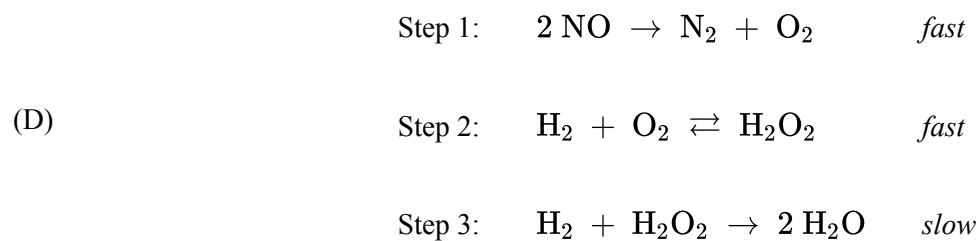
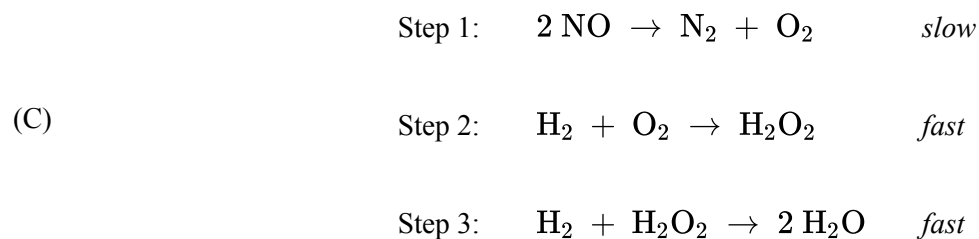
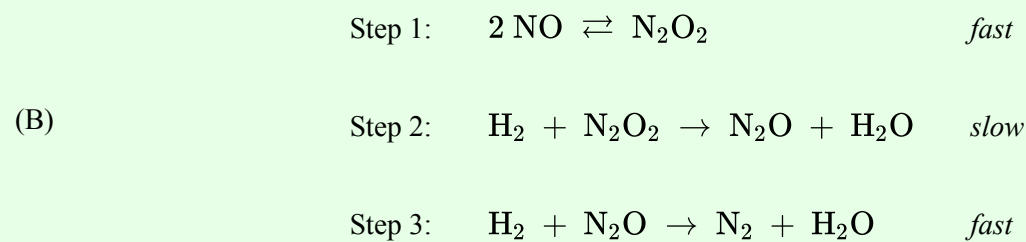
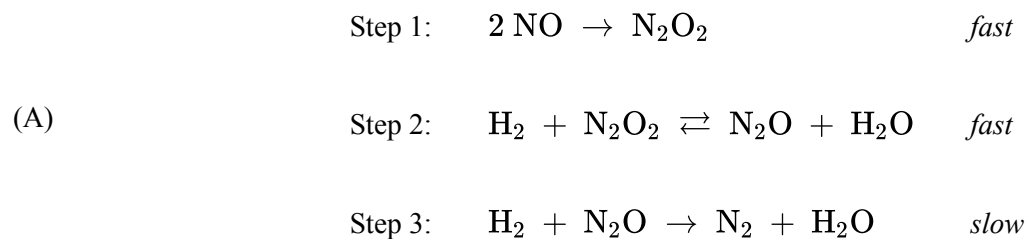
The response provides both the correct answer and a valid justification equivalent to the following.

· No. The linear relationship between $\ln [\text{C}_2\text{H}_6\text{O}]$ versus time is consistent with a first-order reaction, but the slope of the line does not indicate a rate constant of -0.50 min^{-1} . The slope of the line is -0.20 min^{-1} which corresponds to a rate constant of -0.20 min^{-1} , since $\text{slope} = -k$ for a first-order process.



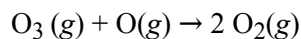
The experimental rate law for the reaction represented above is $\text{rate} = k[\text{H}_2][\text{NO}]^2$. Which of the following proposed mechanisms is consistent with the rate law?

Chapter 09 Part 3: Reaction Mechanism

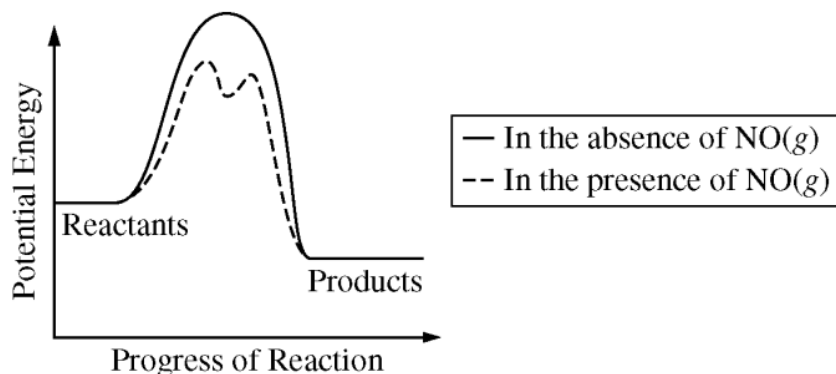


Chapter 09 Part 3: Reaction Mechanism

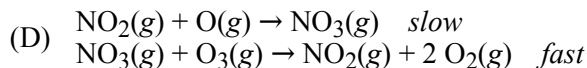
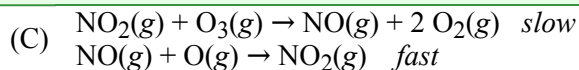
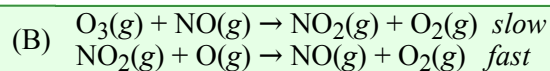
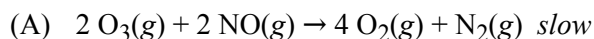
31.



The decomposition of $\text{O}_3(g)$ in the upper atmosphere is represented by the equation above. The potential energy in the upper atmosphere is diagram for the decomposition of $\text{O}_3(g)$ in the presence and absence of $\text{NO}(g)$ is given below.



Which of the following mechanisms for the catalyzed reaction is consistent with the equation and diagram above?



32. $\text{N}_2\text{O}(g) + \text{CO}(g) \rightarrow \text{N}_2(g) + \text{CO}_2(g)$

The rate of the reaction represented above increases significantly in the presence of $\text{Pd}(s)$. Which of the following best explains this observation?

(A) Pd increases the activation energy of the reaction.

(B) Pd absorbs the heat produced in the reaction.

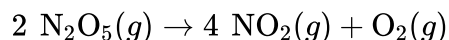
(C) One of the reactants binds on the surface of Pd , which introduces an alternative reaction pathway with a lower activation energy.

(D) One of the products binds on the surface of Pd , which increases the reaction rate by decreasing the concentration of products in the mixture.

Chapter 09 Part 3: Reaction Mechanism

33. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

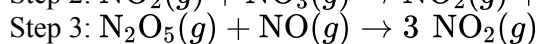
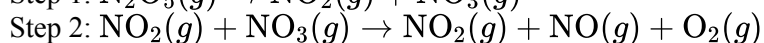
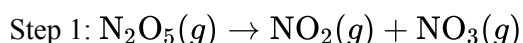
The following equation represents the decomposition of N_2O_5 , for which the rate law is $\text{rate} = k[\text{N}_2\text{O}_5]$.



A sample of pure $\text{N}_2\text{O}_5(g)$ is placed in an evacuated container and allowed to decompose at a constant temperature of 300 K. The concentration of $\text{N}_2\text{O}_5(g)$ in the container is measured over a period of time, and the measurements are recorded in the following table.

Time (hr)	$[\text{N}_2\text{O}_5](M)$
0	0.160
1.67	0.0800
3.33	0.0400
5.00	0.0200

- (a) Determine the value of the rate constant, k , for the reaction. Include units in your answer.
- (b) The following mechanism is proposed for the decomposition of $\text{N}_2\text{O}_5(g)$.



Identify which step of the proposed mechanism (1, 2, or 3) is the rate-determining step. Justify your answer in terms of the rate law given.

- (c) If this experiment was repeated at the same temperature but with twice the initial concentration of N_2O_5 , would the value of k increase, decrease, or remain the same? Explain your reasoning.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1	2
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Chapter 09 Part 3: Reaction Mechanism

The response meets **both** of the following criteria.

- ☐ The response provides the correct calculated value.

$$\circ k = \frac{0.693}{t_{1/2}} = \frac{0.693}{1.67 \text{ hr}} = 0.415 \text{ hr}^{-1}$$

OR

$$\circ k = \frac{\ln[A]_0 - \ln[A]_t}{t} = \frac{\ln(0.160) - \ln(0.0800)}{1.67 \text{ hr}} = 0.415 \text{ hr}^{-1}$$

OR

$$\circ k = \frac{\ln[A]_0 - \ln[A]_t}{t} = \frac{\ln(0.160) - \ln(0.0400)}{3.33 \text{ hr}} = 0.416 \text{ hr}^{-1}$$

OR

$$\circ k = \frac{\ln[A]_0 - \ln[A]_t}{t} = \frac{\ln(0.160) - \ln(0.0200)}{5.00 \text{ hr}} = 0.416 \text{ hr}^{-1}$$

- ☐ The response includes the correct units, consistent with the calculated value.

$$\circ \text{hr}^{-1}$$

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct answer and a valid justification.

- Step 1 is the rate-determining step. The rate law of elementary step 1 is $\text{rate} = k[\text{N}_2\text{O}_5]$, which is consistent with the first order kinetics of the overall rate law.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct answer.

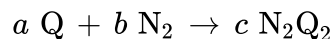
- Remain the same. The rate constant, k , is independent of concentration and will remain the same at constant temperature.

Chapter 09 Part 3: Reaction Mechanism

34. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



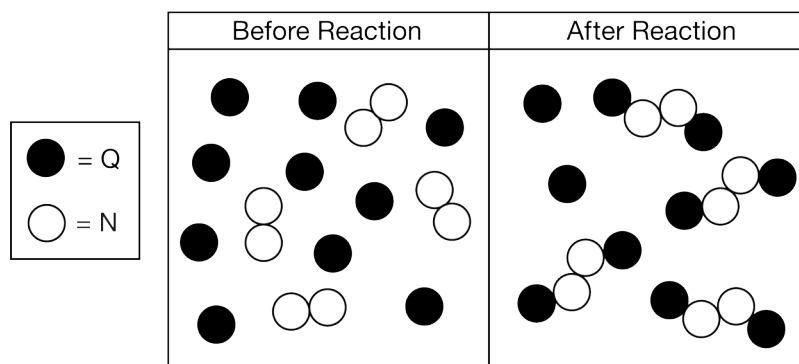
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

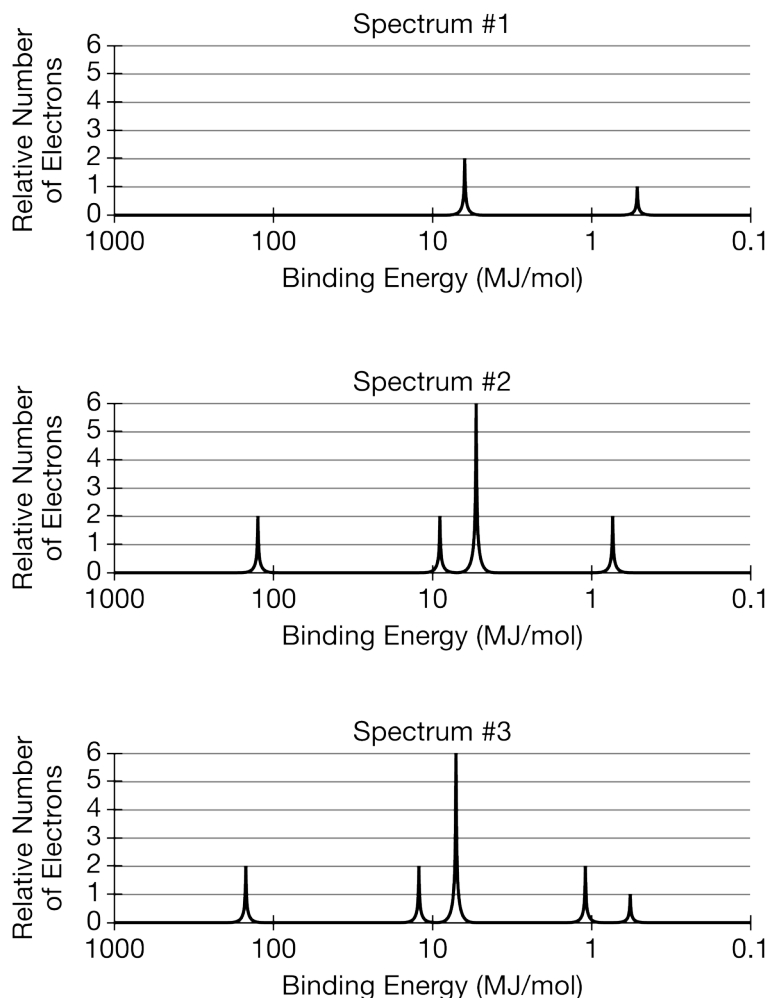


- (b) A student claims that N₂ is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound X₃N.

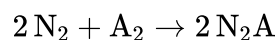
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

Chapter 09 Part 3: Reaction Mechanism



(d) A student claims that the bonds in X_3N are nonpolar covalent. Do you agree or disagree with the student's hypothesis? Justify your choice.

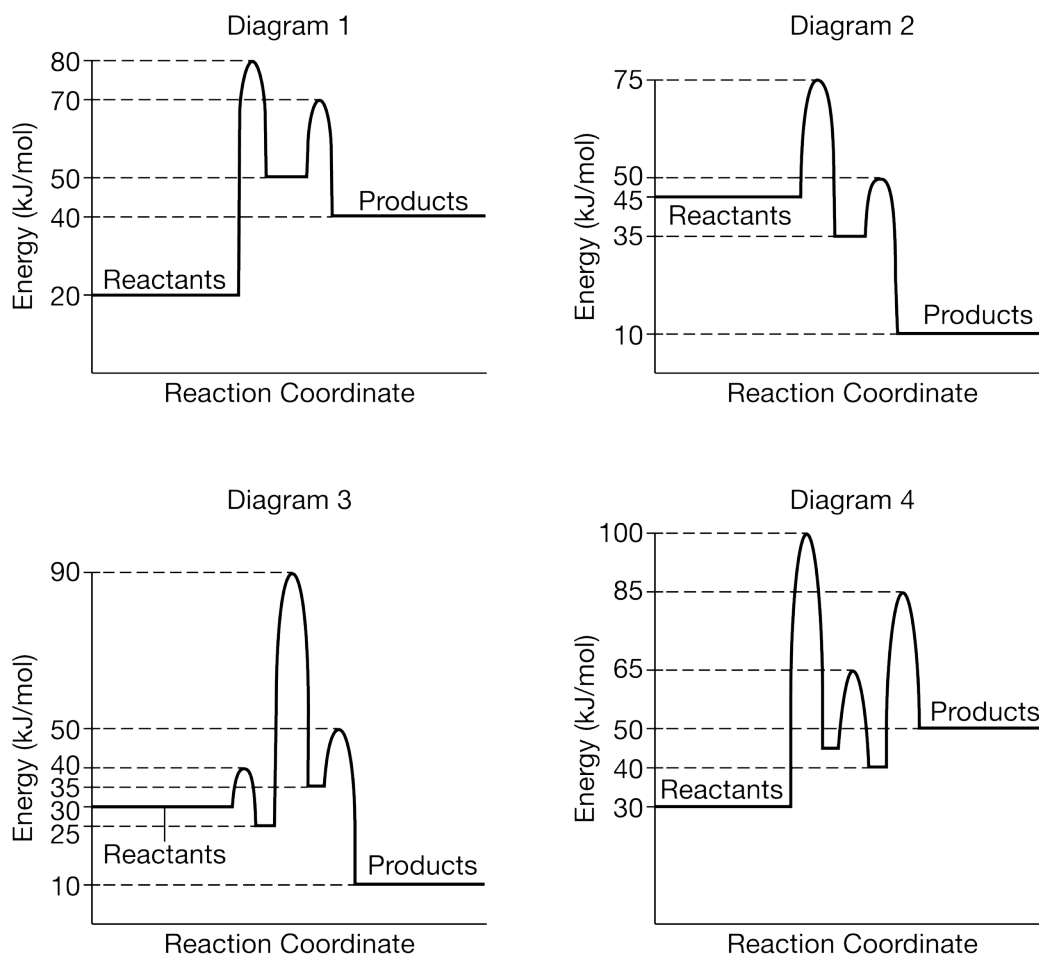
A new diatomic element, A_2 , is discovered and is found to have a molar mass of 52 g/mol. At high temperatures, elemental A_2 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A as shown in the equation below.



(e) Determine the mass of 0.50 mol of N_2A .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ < 0$.

Chapter 09 Part 3: Reaction Mechanism

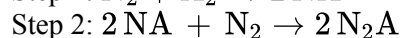
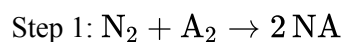


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound NA is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

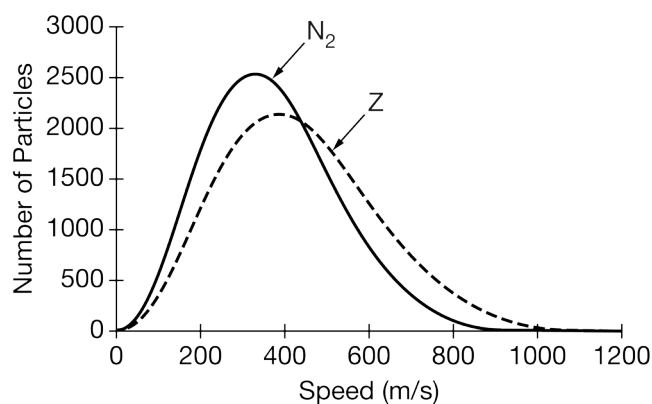
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N ₂	77	5.6
Element R	89	6.8
Element Z	27	1.7

(j) Which sample, 2 moles of N₂ or 10 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, N₂(l) or Z(l), has the stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N₂ and element Z at the same temperature.



(l) Based on the data in the plot, is the molar mass of N₂ greater than, equal to, or less than the molar mass of Z? Justify your answer in terms of kinetic molecular theory.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response includes **all** of the following.

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- ☐ The response shows that $a = 2$.
- ☐ The response shows that $b = 1$.
- ☐ The response shows that $c = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and an explanation equivalent to the following.

- Agree. In the representation, all of the N_2 is consumed in the reaction and some Q remains.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct spectrum choice and a valid justification, based on valence electrons and/or ionic charge, equivalent to the following.

- Spectrum #1. The formula X_3N implies a charge of $+1$ for each ion of X to accompany the -3 charge of the nitride ion. Spectrum #1 is the only spectrum that shows exactly one electron in the valence shell.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on bonding type, equivalent to the following.

- Disagree. The reaction of a metal with a highly electronegative nonmetal (nitrogen) will produce an ionic bond. Nonpolar covalent bonds would be formed between nonmetal elements with similar electronegativities.

Part E

Chapter 09 Part 3: Reaction Mechanism

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct calculation of the mass.

$$\cdot \text{Molar mass of N}_2\text{A} = 28 \text{ g/mol} + \frac{(52 \text{ g/mol})}{2} = 54 \text{ g/mol}$$

$$0.50 \text{ mol N}_2\text{A} \times \frac{54 \text{ g N}_2\text{A}}{1 \text{ mol N}_2\text{A}} = 27 \text{ g N}_2\text{A}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct diagram choice and a valid justification, based on number of peaks and relative position of reactant and product energies, equivalent to the following.

· Diagram 2. The potential energy of the products is lower than that of the reactants, which is consistent with an exothermic reaction. The diagram also has two peaks, corresponding to a two-step mechanism.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides an activation energy that is consistent with the choice in (f).

$$\cdot 75 \text{ kJ/mol} - 45 \text{ kJ/mol} = 30 \text{ kJ/mol}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct step and a justification based on the largest activation energy, consistent with the choice in (f), equivalent to the following.

- The first step. The first step has the greatest activation energy, so it is the slow (rate-determining) step.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

- Disagree. NA is not regenerated at the end of the two-step sequence, so it cannot be a catalyst.

OR

- Disagree. NA is formed in one elementary step of the mechanism and consumed in a subsequent step, making it an intermediate, not a catalyst.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response identifies the correct sample based upon the heat of vaporization and number of moles.

- 10 moles of Z. 10 mol of Z absorbs 17 kJ, while 2 mol of N₂ absorbs only 11.2 kJ.

$$2 \text{ mol N}_2 \times 5.6 \text{ kJ/mol} = 11.2 \text{ kJ}$$

$$10 \text{ mol Z} \times 1.7 \text{ kJ/mol} = 17 \text{ kJ}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets **one** of the following criteria.

Chapter 09 Part 3: Reaction Mechanism

- ☐ The response states that the forces between particles of $\text{N}_2(l)$ must be stronger than those between particles of $\text{Z}(l)$ because the boiling point of N_2 is higher than that of Z .
- ☐ The response states that the forces between particles of $\text{N}_2(l)$ must be stronger than those between particles of $\text{Z}(l)$ because the heat of vaporization of N_2 is greater than that of Z .

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification, based on average speed presented in the plot, equivalent to the following.

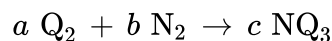
N_2 has a molar mass greater than that of Z . Since the curve for N_2 is shifted further to the left, and the average speed is lower, a particle of N_2 must be more massive than a particle of Z . According to KMT, more massive particles with the same average kinetic energy (temperature) move with lower average speed.

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35. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure diatomic nonmetal, Q_2 , in a closed container containing pure nitrogen gas, N_2 . The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



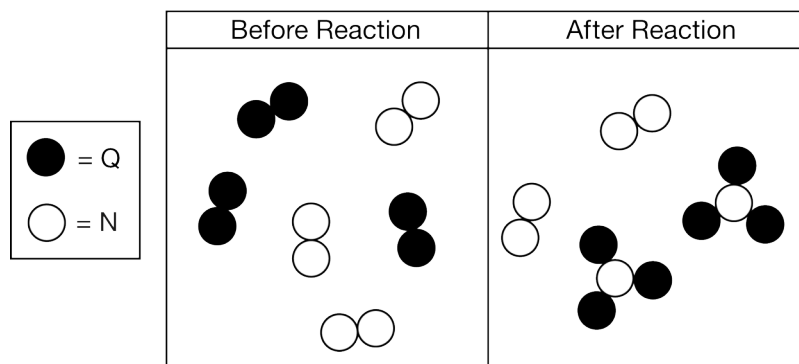
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

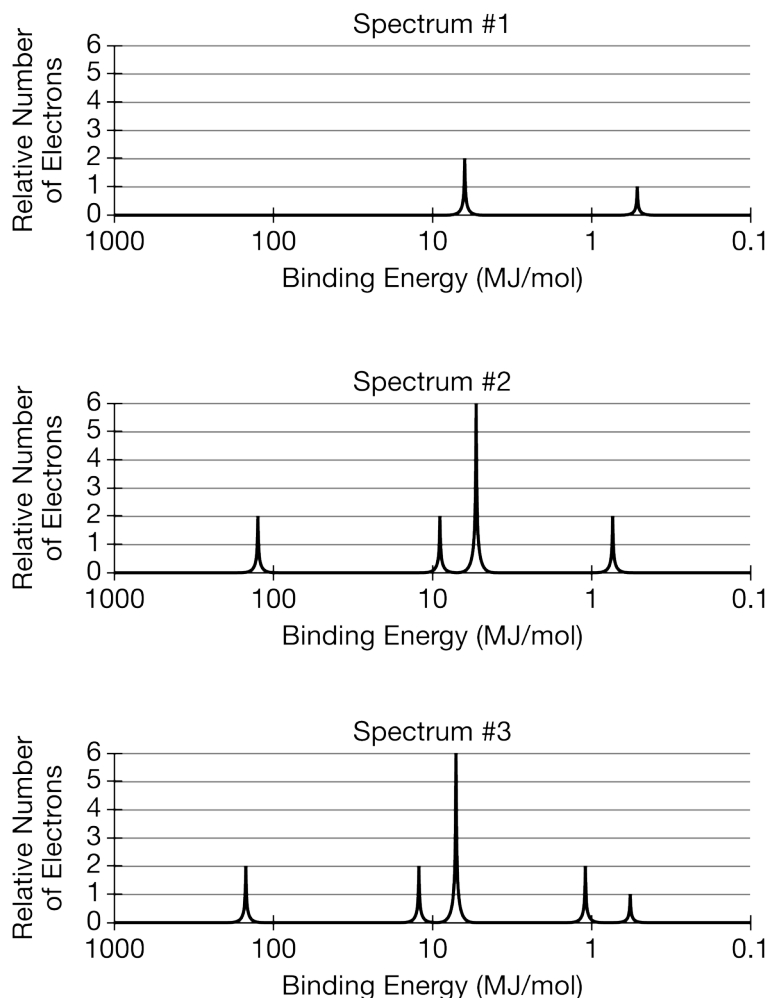


- (b) A student claims that Q_2 is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N_2 reacts with a pure metal X to form the compound X_3N_2 .

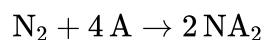
- (c) Which of the following complete photoelectron spectra could be that of metal X ? Justify your choice.

Chapter 09 Part 3: Reaction Mechanism



(d) A student claims that the bonds in X_3N_2 are polar covalent. Do you agree or disagree with the student's claim? Justify your choice.

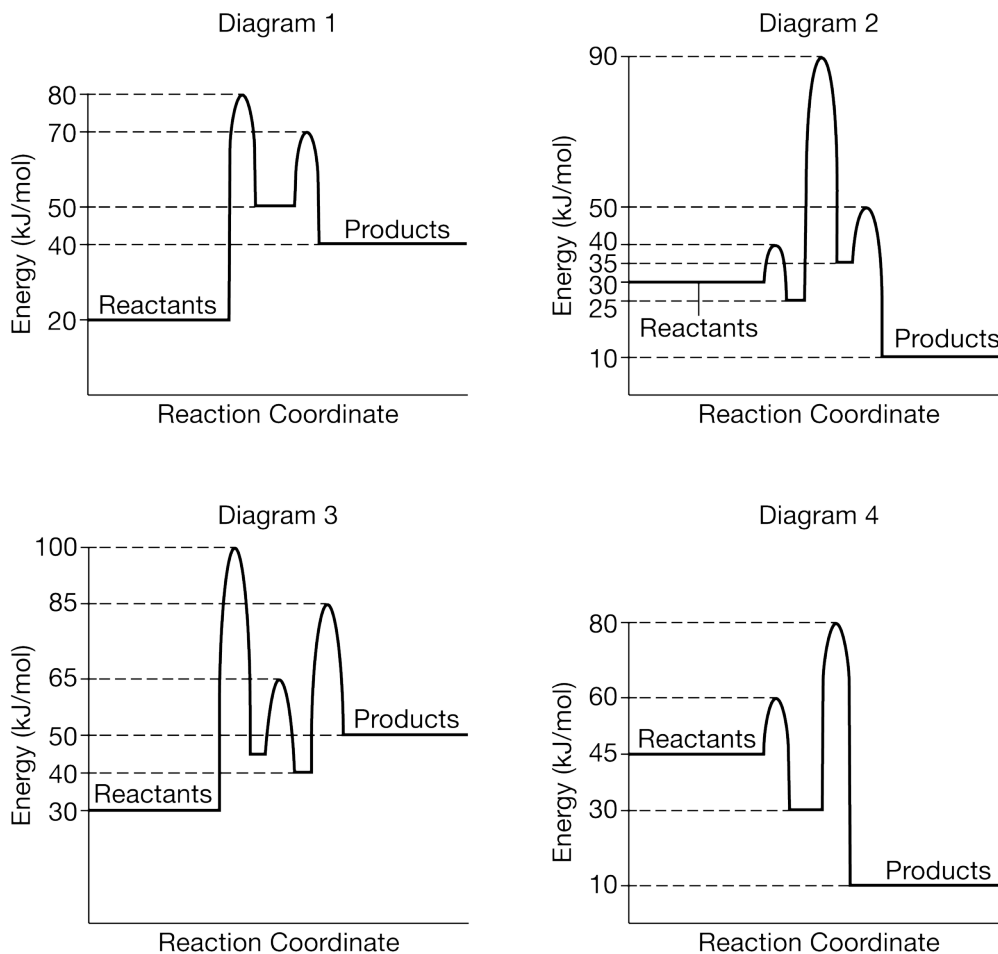
A new monatomic element, A, is discovered and is found to have a molar mass of 18 g/mol. At high temperatures, element A reacts with N_2 (molar mass 28 g/mol) to produce the compound NA_2 as shown in the equation below.



(e) Determine the mass of 0.25 mol of NA_2 .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ < 0$.

Chapter 09 Part 3: Reaction Mechanism

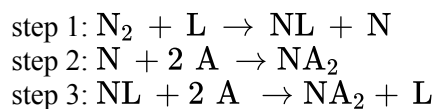


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the element L is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

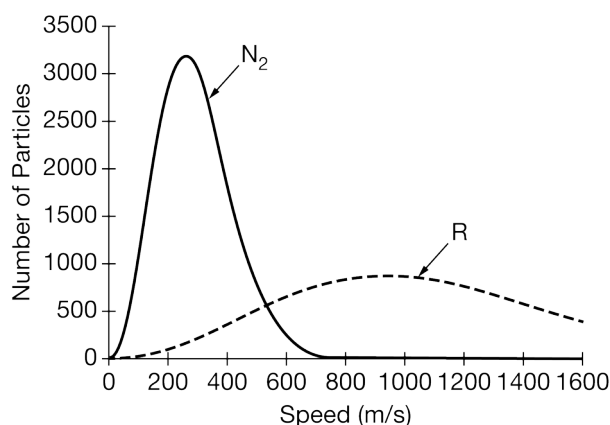
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N ₂	77	5.6
Element R	21	0.45
Element Z	165	12.6

(j) Which sample, 20 moles of R or 3 moles of N₂, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, N₂(l) or R(l), has stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N₂ and element R at the same temperature.



(l) Based on the data in the plot, is the molar mass of R greater than, equal to, or less than the molar mass of N₂? Justify your answer in terms of kinetic molecular theory.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response includes **all** of the following.

Chapter 09 Part 3: Reaction Mechanism

- ☐ The response shows that $a = 3$.
- ☐ The response shows that $b = 1$.
- ☐ The response shows that $c = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives both the correct answer and a valid justification equivalent to the following.

- Agree. In the diagram, all of the Q_2 is consumed in the reaction and some N_2 remains.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct spectrum choice and a valid justification based on valence electrons and/or ionic charge equivalent to the following.

- Spectrum #2. The formula X_3N_2 implies a charge of $+2$ for each ion of X to accompany the -3 charge of the nitride ion. Spectrum #2 is the only spectrum that shows exactly two electrons in the valence shell.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on bonding type, equivalent to the following.

- Disagree. The reaction of a metal with a highly electronegative nonmetal (nitrogen) will produce an ionic bond. Polar covalent bonds would be formed if both elements were nonmetals and had significantly different electronegativities.

Part E

Chapter 09 Part 3: Reaction Mechanism

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct calculation of the mass.

· Molar mass of $\text{NA}_2 = \frac{28 \text{ g/mol}}{2} + 2 \times (18 \text{ g/mol}) = 50 \text{ g/mol}$

$0.25 \text{ mol NA}_2 \times \frac{50 \text{ g NA}_2}{1 \text{ mol NA}_2} = 12.5 \text{ g NA}_2 \approx 13 \text{ g NA}_2$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct diagram choice and a valid justification, based on number of peaks and relative position of reactant and product energies, equivalent to the following.

· Diagram 2. The potential energy of the products is lower than that of the reactants, which is consistent with an exothermic reaction. The diagram also has three peaks, corresponding to a three-step mechanism.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives an activation energy that is consistent with the choice in (f).

· $40 \text{ kJ/mol} - 30 \text{ kJ/mol} = 10 \text{ kJ/mol}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 09 Part 3: Reaction Mechanism

The response provides both the correct step and a justification based on the largest activation energy, consistent with the choice in (f), equivalent to the following.

- The second step. The second step has the greatest activation energy, so it is the slow (rate-determining) step.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- Agree. L is regenerated at the end of the three-step sequence, meaning it is a catalyst.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response identifies the correct sample based upon the heat of vaporization and number of moles.

- 3 moles of N₂ . 3 mol of N₂ absorbs 16.8 kJ, while 20 mol of R absorbs only 9.0 kJ .

$$3 \text{ mol N}_2 \times 5.6 \text{ kJ/mol} = 16.8 \text{ kJ}$$

$$20 \text{ mol R} \times 0.45 \text{ kJ/mol} = 9.0 \text{ kJ}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets **one** of the following criteria.

- ☐ The response states that the intermolecular forces between particles of N₂(l) must be stronger than those between particles of R(l) because the boiling point of N₂ is higher than that of R.
- ☐ The response states that the intermolecular forces between particles of N₂(l) must be stronger than those between particles of R(l) because the heat of vaporization of N₂ is greater than that of R.

Chapter 09 Part 3: Reaction Mechanism**Part L**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification, based on average speed presented in the plot, equivalent to the following.

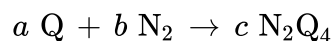
· R has a molar mass less than that of N_2 . Since the curve for R is shifted further to the right, and the average speed is higher, a particle of R must be less massive than a particle of N_2 . According to KMT, less massive particles with the same average kinetic energy (temperature) move with higher average speed.

Chapter 09 Part 3: Reaction Mechanism

36. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



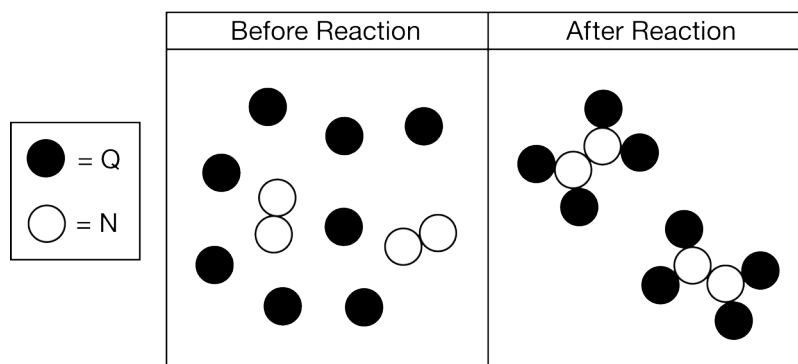
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

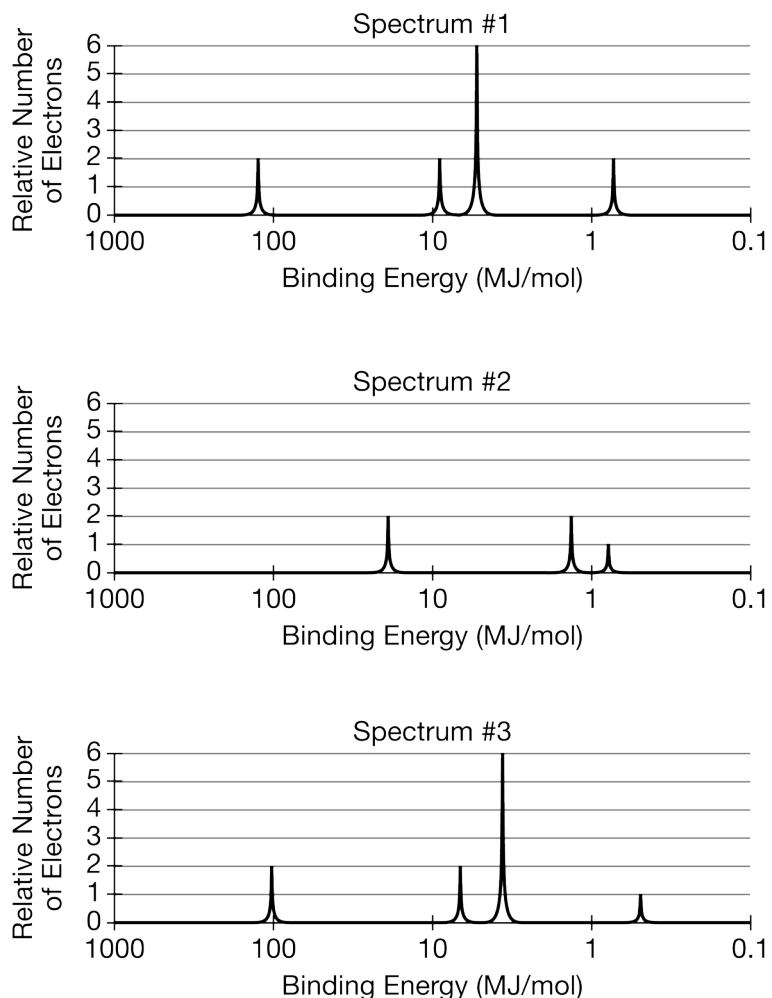


- (b) A student claims that N₂ is the limiting reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound X₃N₂.

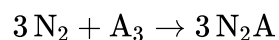
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

Chapter 09 Part 3: Reaction Mechanism



(d) A student claims that molecule X_3N_2 contains both ionic and polar covalent bonds. Do you agree or disagree with the student's claim? Justify your choice.

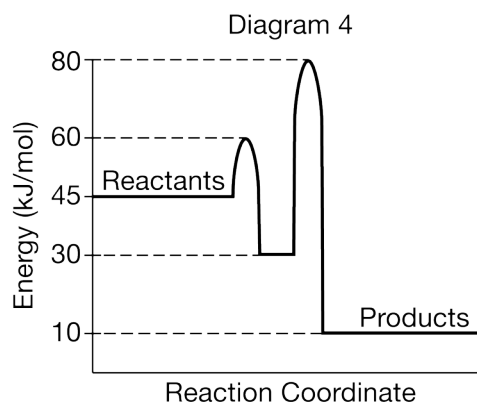
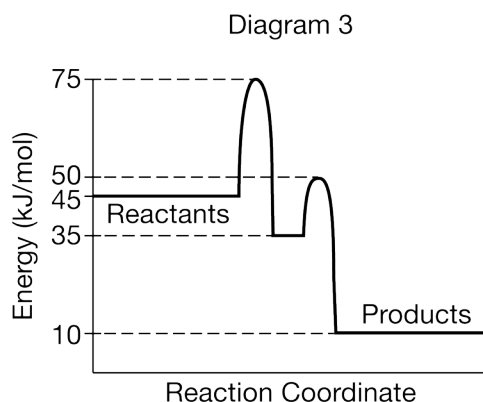
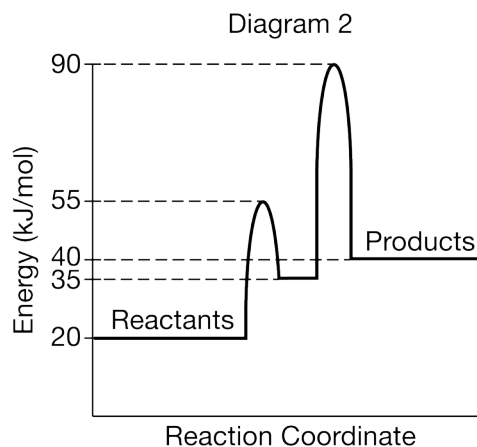
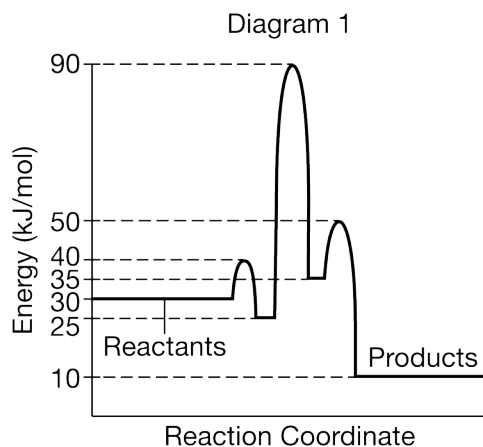
A new triatomic element, A_3 , is discovered and is found to have a molar mass of 93 g/mol. At high temperatures, A_3 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A as shown in the equation below.



(e) Determine the mass of 0.25 mol of N_2A .

The reaction above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ > 0$.

Chapter 09 Part 3: Reaction Mechanism

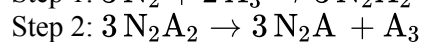
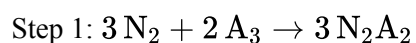


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that element A_3 is an intermediate. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

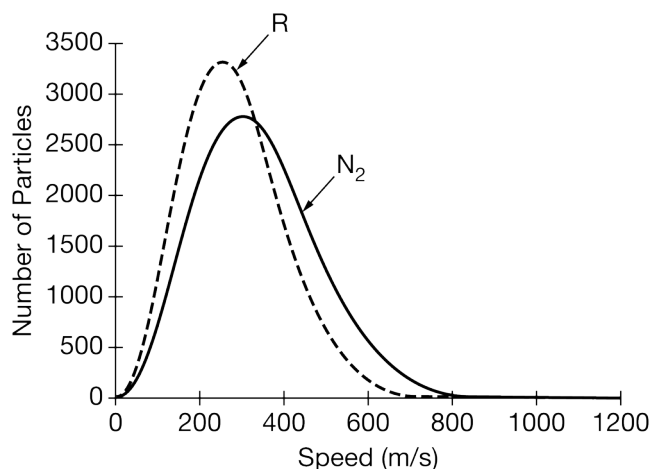
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N_2	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 3 moles of R or 1 mole of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, R(l) or $\text{N}_2(l)$, has weaker attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N_2 and element R at the same temperature.



(l) Based on the data in the plot, is the molar mass of R greater than, equal to, or less than the molar mass of N_2 ? Justify your answer in terms of kinetic molecular theory.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

Chapter 09 Part 3: Reaction Mechanism

The response includes **all** of the following.

- ☐ The response shows that $a = 4$.
- ☐ The response shows that $b = 1$.
- ☐ The response shows that $c = 1$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both a correct claim and a valid justification equivalent to one of the following.

- Agree. In the diagram, the initial amount of N_2 determines the maximum amount of N_2Q_4 that can be produced.

OR

- Disagree or agree. Neither reactant is limiting because N_2 and Q are initially present in the same ratio as in the balanced chemical equation.

OR

- Disagree or agree. Neither reactant is limiting, as no unreacted elements remain.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct spectrum choice and a valid justification, based on valence electrons and/or ionic charge, equivalent to the following.

- Spectrum #1. The formula implies a charge of +2 for each X ion to accompany the -3 charge of the nitride ion. Spectrum #1 is the only spectrum that shows exactly two electrons in the valence shell.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response includes both the correct answer and a valid justification, based on bonding type, equivalent to the following.

· Disagree. The reaction of a metal with a highly electronegative nonmetal (nitrogen) will produce an ionic bond, but there are no polar covalent bonds in X_3N_2 since it is a binary ionic compound.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct calculation of the mass.

· Molar mass of $N_2A = 28 \text{ g/mol} + \frac{(93 \text{ g/mol})}{3} = 59 \text{ g/mol}$

$$0.25 \text{ mol } N_2A \times \frac{59 \text{ g } N_2A}{1 \text{ mol } N_2A} = 14.75 \text{ g } N_2A \approx 15 \text{ g } N_2A$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct diagram choice and a valid justification, based on number of peaks and relative position of reactant and product energies, equivalent to the following.

· Diagram 2. The potential energy of the products is higher than that of the reactants, which is consistent with an endothermic reaction.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism

0	1
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The response provides an activation energy that is consistent with the choice in (f).

· $55 \text{ kJ/mol} - 20 \text{ kJ/mol} = 35 \text{ kJ/mol}$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct step and a justification based on the largest activation energy, consistent with the choice in (f), equivalent to the following.

- The second step. The second step has the greatest activation energy, so it is the slow (rate-determining) step.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

- Disagree. A_3 is regenerated at the end of the two-step sequence, making it a catalyst.

OR

- Disagree. A_3 is not produced in one elementary step of the mechanism and consumed in a subsequent step, so it cannot be an intermediate.

OR

- Disagree. Some A_3 is consumed in the overall reaction, making it a reactant.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response identifies the correct sample based upon the heat of vaporization and number of moles.

· 3 moles of R. 3 mol of R absorbs 20.4 kJ, while 1 mol of Z absorbs only 12.6 kJ.

$$3 \text{ mol R} \times 6.8 \text{ kJ/mol} = 20.4 \text{ kJ}$$

$$1 \text{ mol Z} \times 12.6 \text{ kJ/mol} = 12.6 \text{ kJ}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response meets **one** of the following criteria.

- ☐ The response states that the intermolecular forces between particles of $\text{N}_2(l)$ must be weaker than those between particles of $\text{R}(l)$ because the boiling point of N_2 is lower than that of R.
- ☐ The response states that the intermolecular forces between particles of $\text{N}_2(l)$ must be weaker than those between particles of $\text{R}(l)$ because the heat of vaporization of N_2 is lower than that of R.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

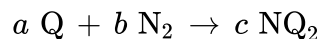
· R has a molar mass greater than that of N_2 . Since the curve for R is shifted further to the left, and the average speed is lower, a particle of R must be more massive than a particle of N_2 . According to KMT, more massive particles with the same average kinetic energy (temperature) move with lower average speed.

Chapter 09 Part 3: Reaction Mechanism

37. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen.

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



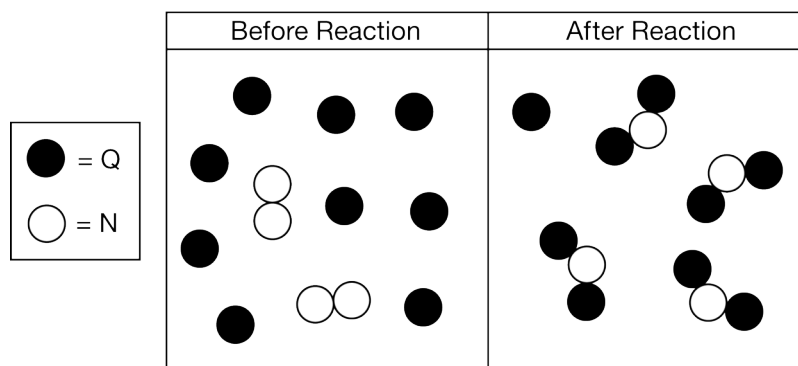
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

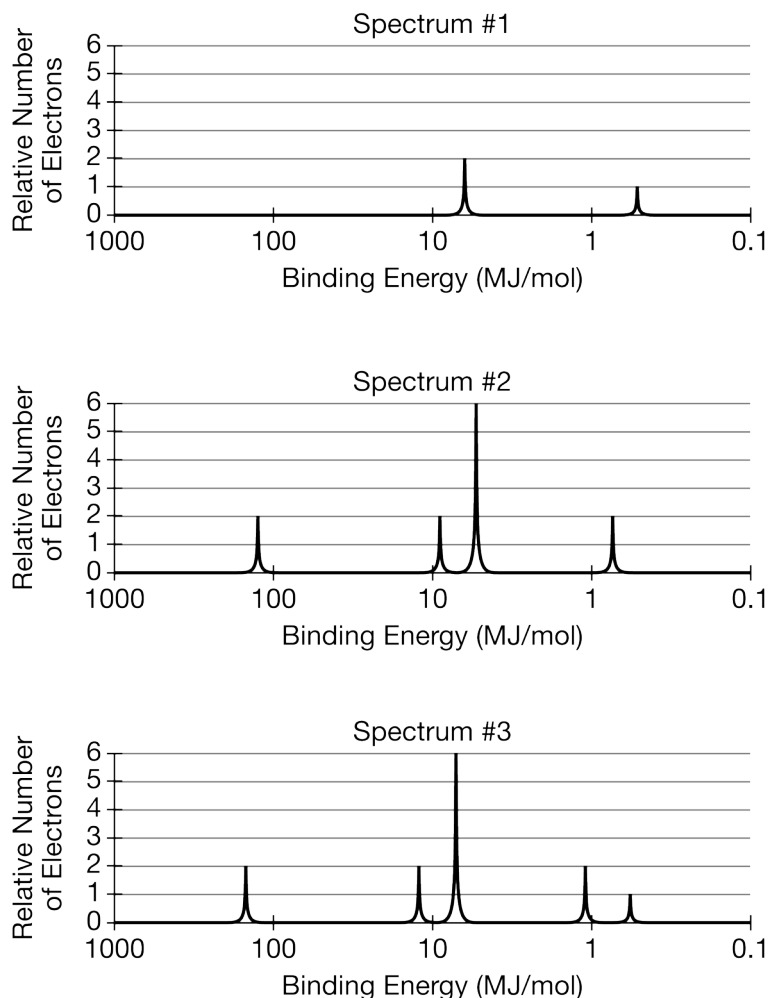


- (b) A student claims that N₂ is the excess reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound XN.

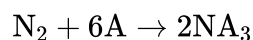
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

Chapter 09 Part 3: Reaction Mechanism



(d) A student claims that the bonds in XN are ionic. Do you agree or disagree with the student's claim? Justify your choice.

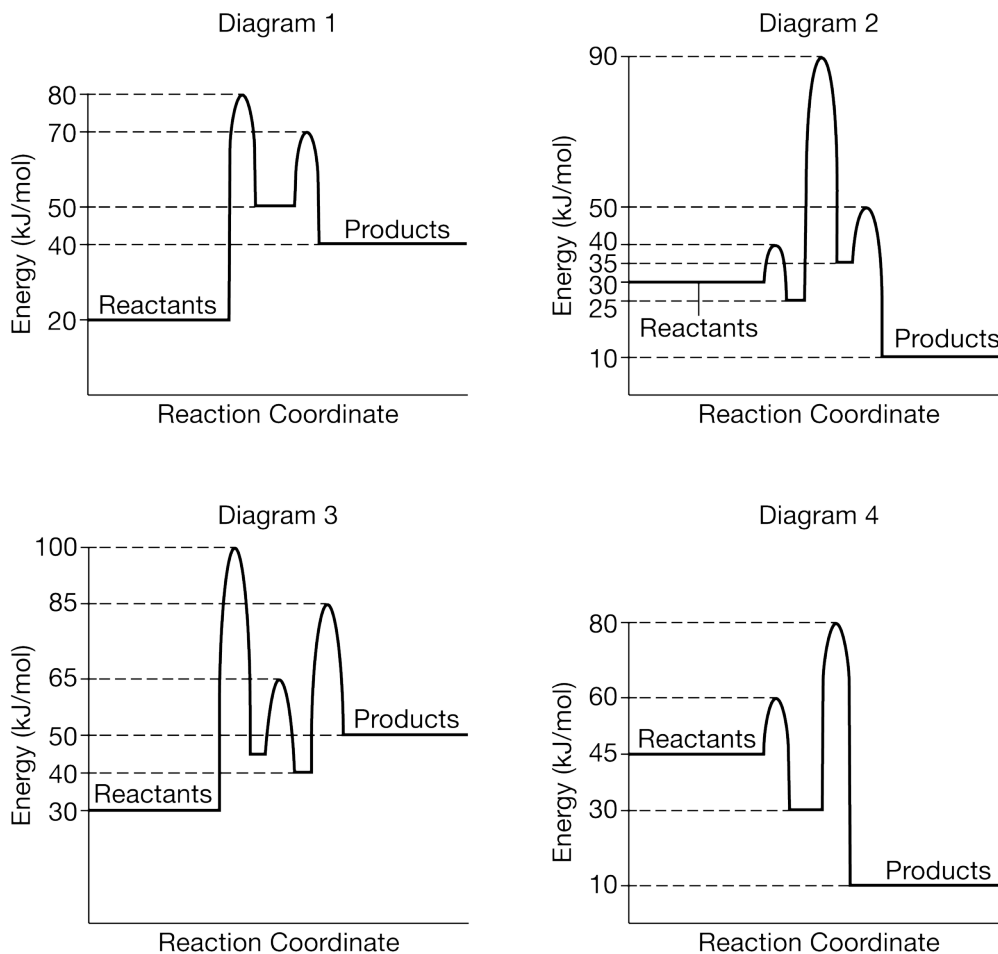
A new monatomic element, A , is discovered and is found to have a molar mass of 13 g/mol . At high temperatures, element A reacts with N_2 (molar mass 28 g/mol) to produce the compound NA_3 as shown in the equation below.



(e) Determine the mass of 0.50 mol of NA_3 .

The reaction described above part (e) occurs via a two-step mechanism and has a $\Delta H_{\text{rxn}}^{\circ} > 0$.

Chapter 09 Part 3: Reaction Mechanism

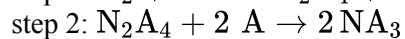
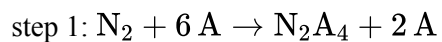


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound N_2A_4 is a catalyst. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

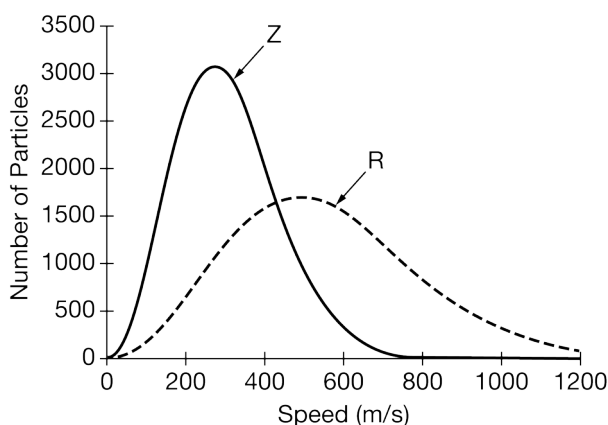
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N ₂	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 5 moles of R or 3 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, Z(l) or R(l), has stronger attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of elements Z and R at the same temperature.



(l) Based on the data in the plot, is the molar mass of Z greater than, equal to, or less than the molar mass of R? Justify your answer in terms of kinetic molecular theory.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response includes **all** of the following.

Chapter 09 Part 3: Reaction Mechanism

- ☐ The response shows that $a = 4$.
- ☐ The response shows that $b = 1$.
- ☐ The response shows that $c = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and an explanation equivalent to the following.

- Disagree. In the diagram, all of the N_2 is consumed in the reaction and some Q remains, making Q the excess reactant.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct spectrum choice and a valid justification, based on valence electrons and/or ionic charge, equivalent to the following.

- Spectrum #3. The formula XN implies a charge of $+3$ for the X ion to accompany the -3 charge of the nitride ion. Spectrum #3 is the only spectrum that shows three electrons in the valence shell.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification, based on bonding type, equivalent to one of the following.

- Agree. The reaction of a metal with a highly electronegative nonmetal (nitrogen) will produce an ionic bond.

OR

- Disagree. Spectrum #3 corresponds to Al, and the compound AlN demonstrates significant covalent character.

Chapter 09 Part 3: Reaction Mechanism**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct calculation of the mass.

$$\cdot \text{Molar mass of } \text{NA}_3 = \frac{(28 \text{ g/mol})}{2} + 3 \times (13 \text{ g/mol}) = 53 \text{ g/mol}$$

$$0.50 \text{ mol NA}_3 \times \frac{53 \text{ g NA}_3}{1 \text{ mol NA}_3} = 26.5 \text{ g} \approx 27 \text{ g NA}_3$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct diagram choice and a valid justification, based on number of peaks and relative position of reactant and product energies, equivalent to the following.

· Diagram 1. The potential energy of the products is higher than that of the reactants, which is consistent with an endothermic reaction. The diagram also has two peaks, corresponding to a two-step mechanism.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides an activation energy that is consistent with the choice in (f).

$$\cdot 80 \text{ kJ/mol} - 20 \text{ kJ/mol} = 60 \text{ kJ/mol}$$

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism

0	1
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The response gives both the correct step and a justification based on the largest activation energy, consistent with the choice in (f), equivalent to the following.

- The first step. The first step has the greatest activation energy, so it is the slow (rate-determining) step.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

- Disagree. N_2A_4 is not regenerated at the end of the two-step sequence, so it cannot be a catalyst.

OR

- Disagree. N_2A_4 is produced in one elementary step of the mechanism and consumed in a subsequent step, making it an intermediate, not a catalyst.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response identifies the correct sample based upon the heat of vaporization and number of moles.

- 3 moles of Z. 3 mol of Z absorbs 37.8 kJ, while 5 mol of R absorbs only 34 kJ.

$$5 \text{ mol R} \times 6.8 \text{ kJ/mol} = 34 \text{ kJ}$$

$$3 \text{ mol Z} \times 12.6 \text{ kJ/mol} = 37.8 \text{ kJ}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response meets **one** of the following criteria.

- ☐ The response states that the intermolecular forces between particles of $Z(l)$ must be stronger than those between particles of $R(l)$ because the boiling point of Z is higher than that of R.
- ☐ The response states that the intermolecular forces between particles of $Z(l)$ must be stronger than those between particles of $R(l)$ because the heat of vaporization of Z is greater than that of R.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives both the correct answer and a valid justification, based on average speed presented in the plot, equivalent to the following.

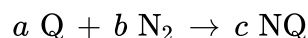
· Z has a molar mass greater than that of R. Since the curve for Z is shifted further to the left, and the average speed is lower, a particle of Z must be more massive than a particle of R. According to KMT, more massive particles with the same average kinetic energy (temperature) move with lower average speed.

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38. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions about some of the properties and reactions of nitrogen

A scientist heats a sample of a pure nonmetal, Q, in a closed container containing pure nitrogen gas, N₂. The reaction proceeds to completion. The unbalanced chemical equation for the reaction appears below.



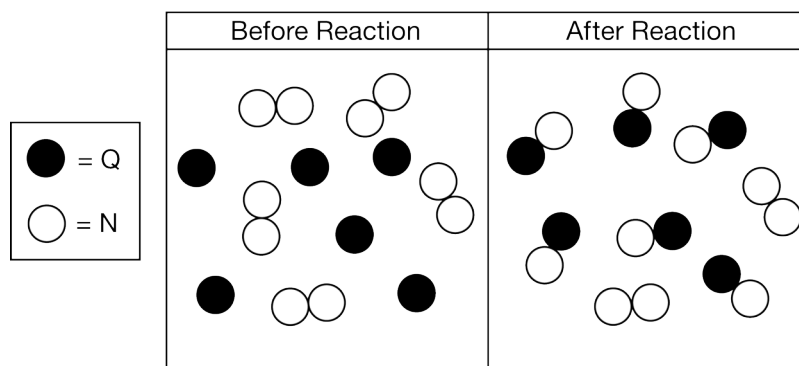
- (a) Give the correct value of each stoichiometric coefficient for the reaction.

$a =$

$b =$

$c =$

A particulate representation from before and after the reaction is shown below. The contents of the container before the reaction are represented in the box on the left, and the contents of the container after the reaction are shown in the box on the right.

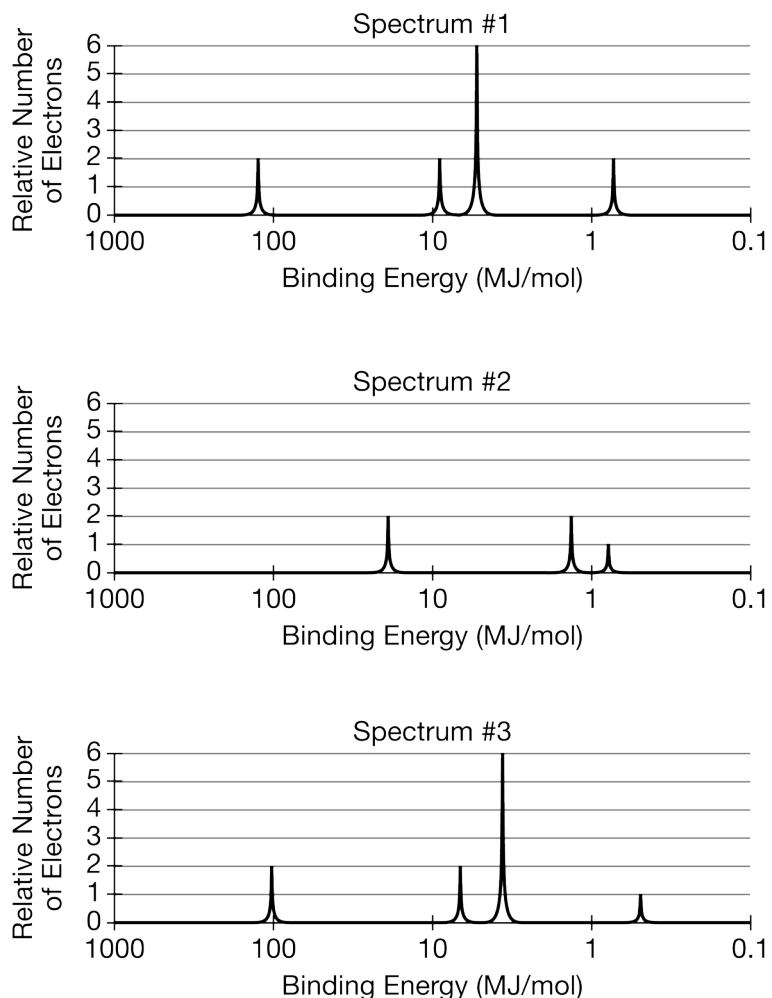


- (b) A student claims that Q is the excess reactant. Do you agree or disagree? Justify your answer.

In a different reaction, N₂ reacts with a pure metal X to form the compound X₃N.

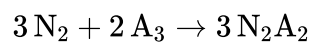
- (c) Which of the following complete photoelectron spectra could be that of metal X? Justify your choice.

Chapter 09 Part 3: Reaction Mechanism



(d) A student claims that molecule X_3N contains both ionic and nonpolar covalent bonds. Do you agree or disagree with the student's claim? Justify your choice.

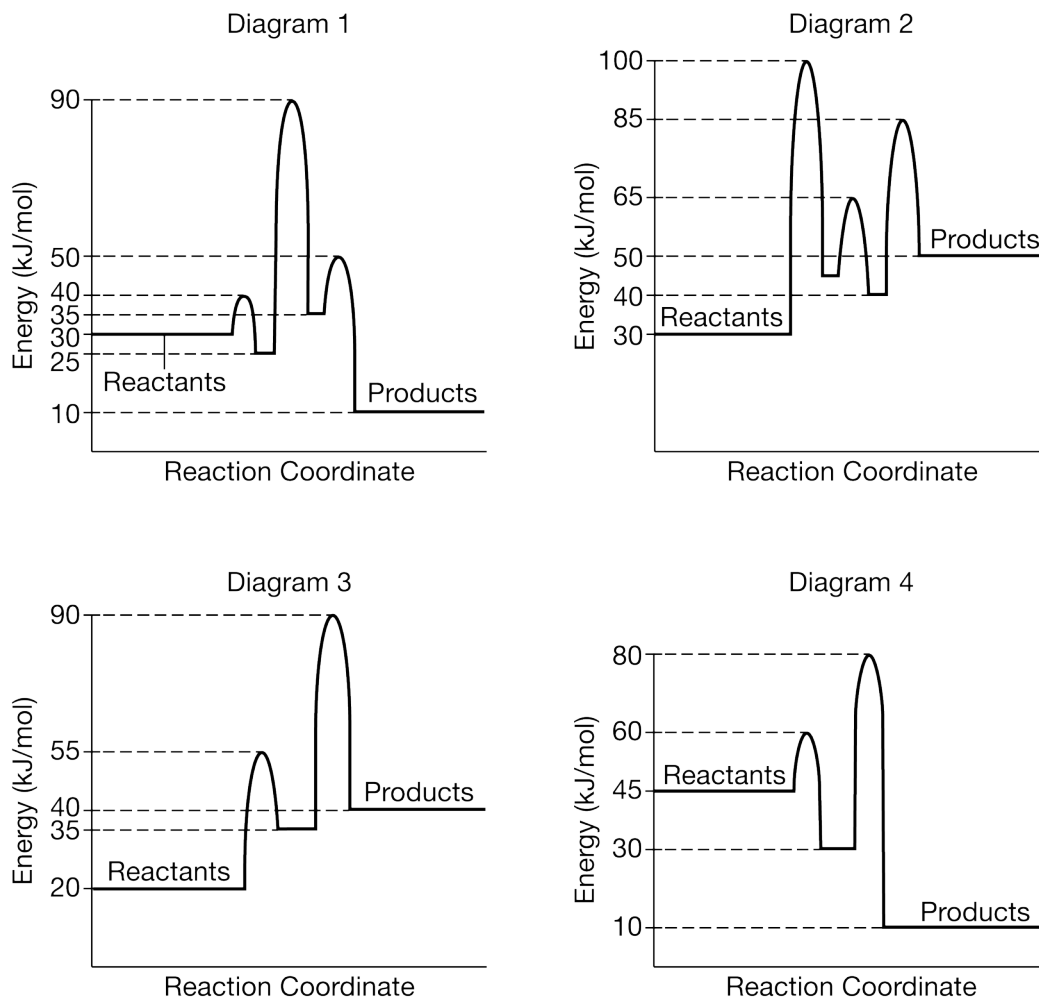
A new triatomic element, A_3 , is discovered and is found to have a molar mass of 45 g/mol. At high temperatures, A_3 reacts with N_2 (molar mass 28 g/mol) to produce the compound N_2A_2 as shown in the equation below.



(e) Determine the mass of 0.50 mol of N_2A_2 .

The reaction above part (e) occurs via a two-step mechanism and has a $\Delta H_{rxn}^\circ > 0$.

Chapter 09 Part 3: Reaction Mechanism

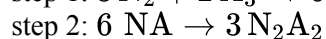
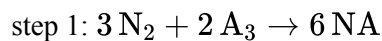


(f) Identify which of the diagrams shown above could represent this reaction. Justify your choice.

(g) Using the diagram selected in part (f), determine the activation energy of the first step of the reaction mechanism in kJ/mol.

(h) Identify the rate-determining step in the diagram selected in part (f). Justify your choice.

A proposed mechanism for the reaction is shown below.



(i) A student claims that the compound NA is an intermediate. Do you agree or disagree with the student's claim? Explain your choice.

The table below lists the properties of three nonmetals.

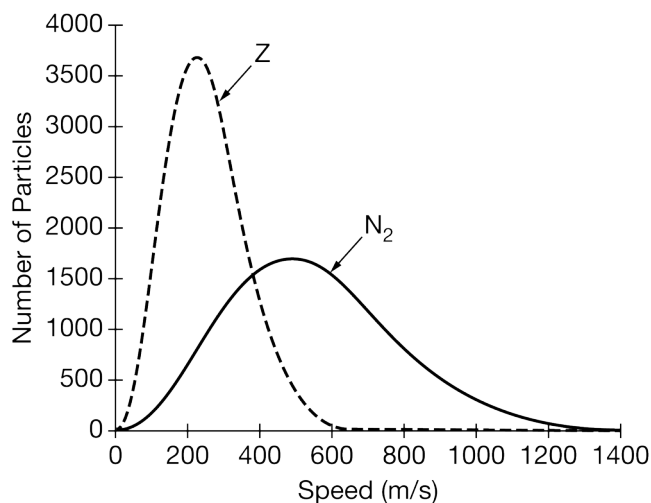
Chapter 09 Part 3: Reaction Mechanism

Substance	Boiling Point (K)	Heat of Vaporization (kJ/mol)
Nitrogen, N ₂	77	5.6
Element R	89	6.8
Element Z	165	12.6

(j) Which sample, 5 moles of N₂ or 2 moles of Z, requires the greatest amount of energy to vaporize at its boiling point? Justify your answer using information in the table.

(k) Which substance, Z(l) or N₂(l), has weaker attractions among its particles? Justify your answer using information from the table.

The plot below shows the distribution of particle speeds for equimolar amounts of N₂ and element Z at the same temperature.



(l) Based on the data in the plot, is the molar mass of Z greater than, equal to, or less than the molar mass of N₂? Justify your answer in terms of kinetic molecular theory.

Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response includes **all** of the following.

- ☐ The response shows that $a = 2$.
- ☐ The response shows that $b = 1$.
- ☐ The response shows that $c = 2$.

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct answer and an explanation equivalent to the following.

- Disagree. In the diagram, all of the Q is consumed in the reaction and some N_2 remains, making N_2 the excess reactant.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes the correct spectrum choice and a valid justification based on valence electrons and/or ionic charge equivalent to the following.

- Spectrum #3. The formula implies a charge of $+1$ for each X ion to accompany the -3 charge of the nitride ion. Spectrum #3 is the only spectrum that shows exactly one electron in the valence shell.

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 09 Part 3: Reaction Mechanism

The response provides both the correct answer and a valid justification, based on bonding type, equivalent to the following.

· Disagree. The reaction of a metal with a highly electronegative nonmetal (nitrogen) will produce an ionic bond, but there are no nonpolar covalent bonds in X_3N since it is a binary ionic compound.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct calculation of the mass.

· Molar mass of $N_2A_2 = 28 \text{ g/mol} + \frac{2}{3}(45 \text{ g/mol}) = 58 \text{ g/mol}$

$$0.50 \text{ mol } N_2A_2 \times \frac{58 \text{ g } N_2A_2}{1 \text{ mol } N_2A_2} = 29 \text{ g } N_2A_2$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response includes both the correct diagram choice and a valid justification, based on number of peaks and relative position of reactant and product energies, equivalent to the following.

· Diagram 3. The potential energy of the products is higher than that of the reactants, which is consistent with an endothermic reaction. The diagram also has two peaks, corresponding to a two-step mechanism.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides an activation energy that is consistent with the choice in (f).

· $55 \text{ kJ/mol} - 20 \text{ kJ/mol} = 35 \text{ kJ/mol}$

Part H

Chapter 09 Part 3: Reaction Mechanism

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct step and a justification based on the largest activation energy, consistent with the choice in (f), equivalent to the following.

- The second step. The second step has the greatest activation energy, so it is the slow (rate-determining) step.

Part I

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- Agree. NA is produced in one elementary step of the mechanism and consumed in a subsequent step, making it an intermediate.

Part J

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response identifies the correct sample based upon the heat of vaporization and number of moles.

- 5 moles of N_2 . 5 mol of N_2 absorbs 28 kJ, while 2 mol of Z absorbs only 25.2 kJ.

$$5 \text{ mol N}_2 \times 5.6 \text{ kJ/mol} = 28 \text{ kJ}$$

$$2 \text{ mol Z} \times 12.6 \text{ kJ/mol} = 25.2 \text{ kJ}$$

Part K

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response meets **one** of the following criteria.

- ☐ The response states that the intermolecular forces between particles of $\text{N}_2(l)$ must be weaker than those between particles of $\text{Z}(l)$ because the boiling point of N_2 is lower than that of Z .
- ☐ The response states that the intermolecular forces between particles of $\text{N}_2(l)$ must be weaker than those between particles of $\text{Z}(l)$ because the heat of vaporization of N_2 is lower than that of Z .

Part L

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



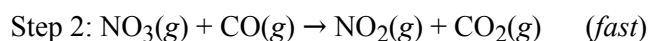
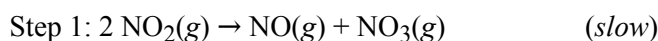
0	1
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The response provides both the correct answer and a valid justification, based on average speed presented in the plot, equivalent to the following.

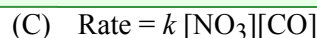
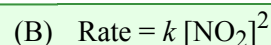
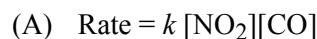
$\cdot$ Z has a molar mass greater than that of N_2 . Since the curve for Z is shifted further to the left, and the average speed is lower, a particle of Z must be more massive than a particle of N_2 . According to **KMT**, more massive particles with the same average kinetic energy (temperature) move with lower average speed.



The reaction between $\text{NO}_2(g)$ and $\text{CO}(g)$ is represented above. The elementary steps of a proposed reaction mechanism are represented below.



Which of the following is the rate law for the overall reaction that is consistent with the proposed mechanism?



40.

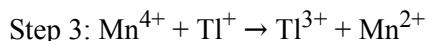
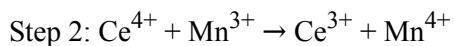
Step 1: ? (*slow*)

Step 2: $\text{NO}_3(g) + \text{CO}(g) \rightarrow \text{NO}_2(g) + \text{CO}_2(g)$ (fast)

Overall: $\text{NO}_2(g) + \text{CO}(g) \rightarrow \text{NO}(g) + \text{CO}_2(g)$

A two-step reaction mechanism is proposed for a gas-phase reaction, as represented above. Which of the following correctly identifies both the chemical equation for step 1 and the rate law for the overall reaction?

- (A) The chemical equation for step 1 is $2 \text{NO}_2(g) \rightarrow \text{NO}(g) + \text{NO}_3(g)$, and the rate law is $rate = [\text{NO}_2]^2$.
- (B) The chemical equation for step 1 is $\text{NO}_2(g) + \text{CO}(g) \rightarrow \text{NO}(g) + \text{CO}_2(g)$, and the rate law is $rate = [\text{NO}_2][\text{CO}]$.
- (C) The chemical equation for step 1 is $\text{NO}_3(g) + 2 \text{CO}(g) \rightarrow \text{NO}(g) + 2 \text{CO}_2(g)$, and the rate law is $rate = [\text{NO}_2][\text{CO}]$.
- (D) The chemical equation for step 1 is $\text{NO}_3(g) + \text{NO}_2(g) + 2 \text{CO}(g) \rightarrow \text{NO}(g) + \text{NO}_2(g) + 2 \text{CO}_2(g)$, and the rate law is $rate = [\text{NO}_3][\text{NO}_2][\text{CO}]^2$.



The proposed steps for a catalyzed reaction between Ce^{4+} and Ti^{+} are represented above. The products of the overall catalyzed reaction are

(A) Ce^{4+} and Tl^{+}

(B) Ce^{3+} and Tl^{3+}

(C) Ce^{3+} and Mn^{3+}

(D) Ce^{3+} and Mn^{4+}

(E) Tl^{3+} and Mn^{2+}

Chapter 09 Part 3: Reaction Mechanism

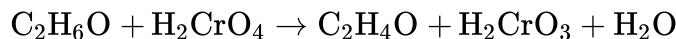
42. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, Q and X, are shown in the following table.

Compound	Q	X
Diagram	$\begin{array}{c} \text{H} \quad \text{H} \\ \quad \\ \text{H}-\text{C}\equiv\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \\ \text{H} \quad \text{H} \end{array}$	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad :\text{O}: \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{C}-\ddot{\text{O}}-\text{H} \\ \quad \quad \quad \uparrow \\ \text{H} \quad \text{H} \quad \text{H} \end{array}$
Boiling Point	281 K	437 K

- (a) Identify the hybridization of the valence orbitals of the carbon atom in compound X that is indicated by the arrow in the diagram.
- (b) The C — H bonds in compound Q are shorter than the C — C bonds in compound X. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound Q –
- Compound X –
- (d) Explain why the boiling point of compound X is higher than that of compound Q based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound Q is placed in an evacuated cylinder with a movable piston that can change the volume of the cylinder. The initial volume and temperature of the gas are 10.0 L and 400. K. If the temperature of the gas is raised to 500. K as its pressure remains constant, what is the final volume of the gas?

Another carbon-containing compound is $\text{C}_2\text{H}_6\text{O}$, which reacts with H_2CrO_4 , as represented by the following equation.



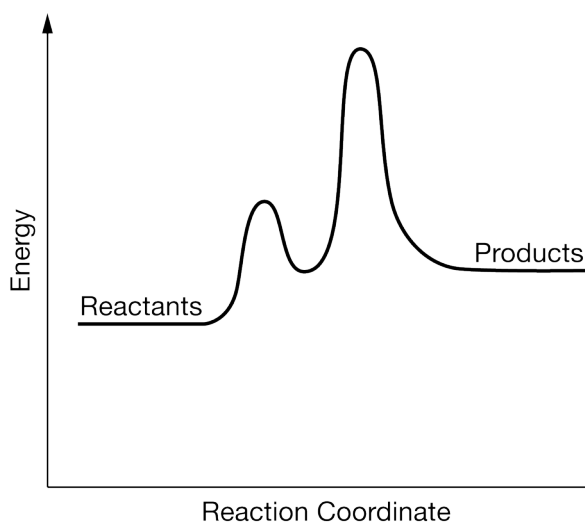
- (f) The student claims that Cr is reduced when H_2CrO_4 is converted to H_2CrO_3 . Do you agree or disagree? Justify your answer in terms of oxidation numbers.

Chapter 09 Part 3: Reaction Mechanism

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

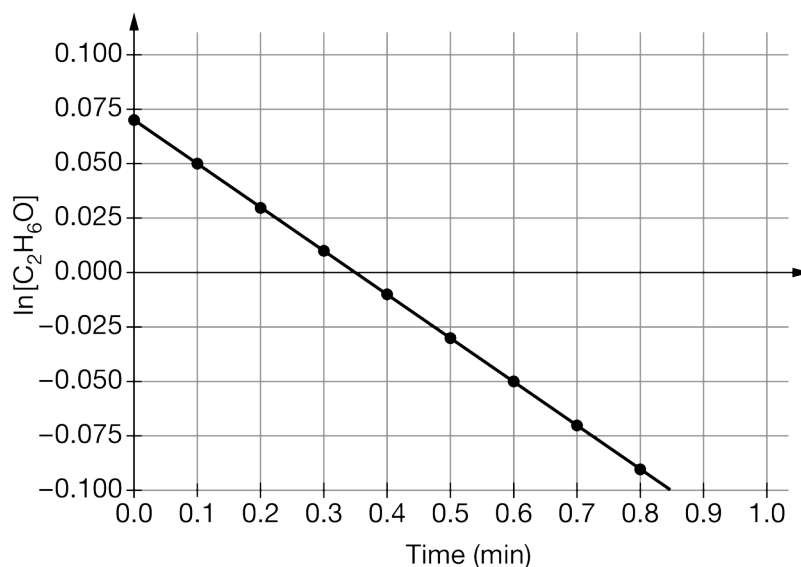
Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is second-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is $0.20 \text{ M}^{-1} \text{ min}^{-1}$. Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.

Chapter 09 Part 3: Reaction Mechanism



Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides the correct hybridization.

· sp^2

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

The response provides a valid explanation equivalent to the following.

· The atomic radius of hydrogen is smaller than that of carbon because the valence electrons of hydrogen occupy a lower shell than those of carbon ($n = 1$ versus $n = 2$). Thus, the internuclear distance (bond length) is shorter for the C – H bonds than for the C – C bonds.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism

0	1
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The response correctly identifies the intermolecular forces in both compounds.

- Compound Q – dispersion forces
- Compound X – dispersion forces, dipole-dipole forces, and hydrogen bonding

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct explanation equivalent to the following.

· The greater number of atoms in compound X leads to a larger, more polarizable electron cloud than in compound Q, and correspondingly stronger London dispersion forces. Compound X also has relatively strong dipole-dipole forces and hydrogen bonding. Because the total of all intermolecular forces in compound X is greater than that in compound Q, compound X has the higher boiling point.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct calculation.

$$\cdot \frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{10.0 \text{ L}}{400. \text{ K}} = \frac{V_2}{500. \text{ K}}$$

$$V_2 = 12.5 \text{ L}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

Chapter 09 Part 3: Reaction Mechanism



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

- Agree. The oxidation number of chromium changes from +6 to +4, indicating that it is reduced.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides correct answer and a valid explanation equivalent to the following.

- No. The activation energies indicated by the relative heights of the peaks are consistent with the second step being rate-determining, but the relative position of the reactants and products is incorrect. The higher energy of the products relative to the reactants indicates an endothermic reaction, not an exothermic reaction.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

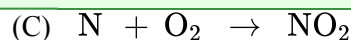
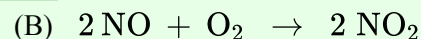
- No. The linear relationship between $\ln [\text{C}_2\text{H}_6\text{O}]$ vs. time is consistent with a first order reaction, not a second order reaction.

OR

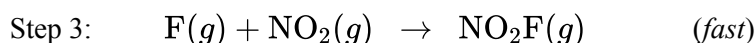
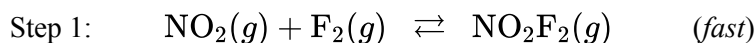
- No. The slope of the line is -0.20 min^{-1} , which is inconsistent with both the mathematical sign and units of a second order process. A second order reaction would give a line with a positive slope (equal to k) and have a rate constant with units of $M^{-1}\text{min}^{-1}$, rather than min^{-1} .

43. $\text{NO}(g)$ and $\text{O}_2(g)$ react to form $\text{NO}_2(g)$. The rate law of the reaction is $rate = k[\text{NO}]^2[\text{O}_2]$. If the reaction occurs in a single elementary step that is a three-body molecular collision, then which of the following is the equation for the elementary step?

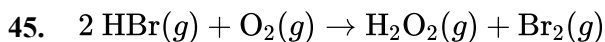
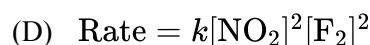
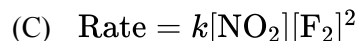
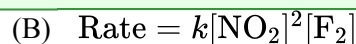
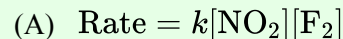
Chapter 09 Part 3: Reaction Mechanism



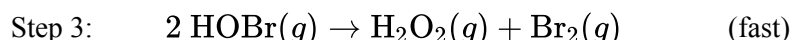
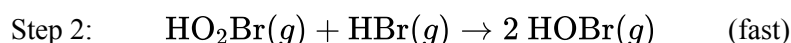
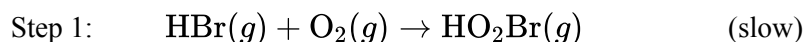
44.



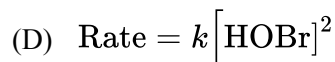
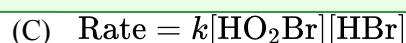
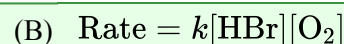
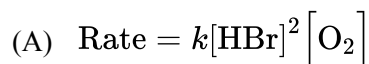
A proposed mechanism for the chemical reaction $2\text{NO}_2(g) + \text{F}_2(g) \rightarrow 2\text{NO}_2\text{F}(g)$ is shown above. Which of the following rate laws is consistent with this mechanism?



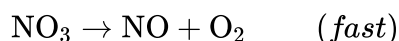
Based on a kinetics study of the reaction represented by the equation above, the following mechanism for the reaction is proposed.



Which of the following rate laws is consistent with the proposed mechanism?



Chapter 09 Part 3: Reaction Mechanism



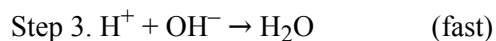
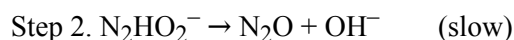
A proposed two-step mechanism for the reaction $2 \text{NO}_2 \rightarrow 2 \text{NO} + \text{O}_2$ is represented. Which of the following rate laws is consistent with the mechanism?

(A) $\text{rate} = k[\text{NO}_3]$

(B) $\text{rate} = k[\text{NO}_2]^2$

(C) $\text{rate} = k[\text{NO}_2]^2[\text{NO}_3]$

(D) $\text{rate} = k \frac{[\text{NO}]^2[\text{O}_2]}{[\text{NO}_2]^2}$



Nitramide, $\text{N}_2\text{H}_2\text{O}_2$, decomposes slowly in aqueous solution. This decomposition is believed to occur according to the reaction mechanism above. The rate law for the decomposition of nitramide that is consistent with this mechanism is given by which of the following?

(A) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2]$

(B) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2][\text{H}^+]$

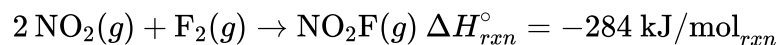
(C) $\text{Rate} = k \frac{[\text{N}_2\text{H}_2\text{O}_2]}{[\text{H}^+]}$

(D) $\text{Rate} = k \frac{[\text{N}_2\text{H}_2\text{O}_2]}{[\text{N}_2\text{HO}_2]}$

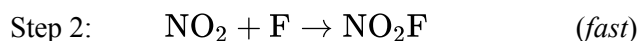
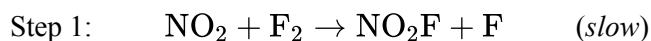
(E) $\text{Rate} = k[\text{N}_2\text{H}_2\text{O}_2][\text{OH}^-]$

Chapter 09 Part 3: Reaction Mechanism

48. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



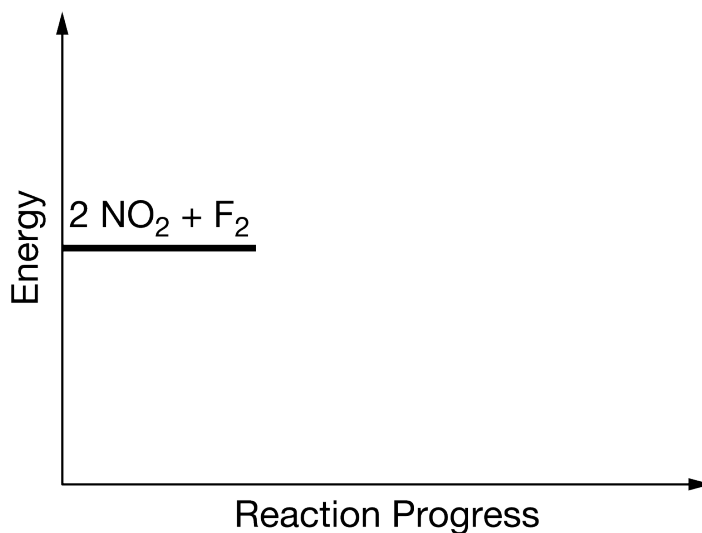
$\text{NO}_2(g)$ and $\text{F}_2(g)$ can react to produce $\text{NO}_2\text{F}(g)$, as represented above. A proposed mechanism for the reaction has two elementary steps, as shown below.



- (a) Write a rate law for the overall reaction that is consistent with the proposed mechanism.
- (b) On the incomplete reaction energy diagram below, draw a curve that shows the following two details.

The relative activation energies of the two elementary steps

The enthalpy change of the overall reaction

**Part (a)**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0

1

Chapter 09 Part 3: Reaction Mechanism

The response gives the following rate law:

$$\text{Rate} = k[\text{NO}_2][\text{F}_2]$$

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

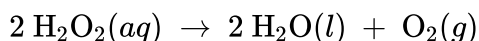


0	1	2	3
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The drawn plot meets all three of the following criteria:

- ☐ There is a curve that starts at the right end of the reactant line, goes up to form one or more peaks, and then goes down and ends at the level of the energy of the product.
- ☐ The curve has two peaks, the first higher than the second.
- ☐ The drawn plot has a horizontal line representing the energy of the product that is lower than the line for the reactants.

The following balanced equation represents the decomposition of hydrogen peroxide, H_2O_2 .



A student carries out an experiment in which a sample of $\text{H}_2\text{O}_2(aq)$ decomposes in the presence of $\text{MnO}_2(s)$. The student measures the volume of $\text{O}_2(g)$ produced by the reaction at regular intervals while the temperature is held constant. The student's data are given in the following table.

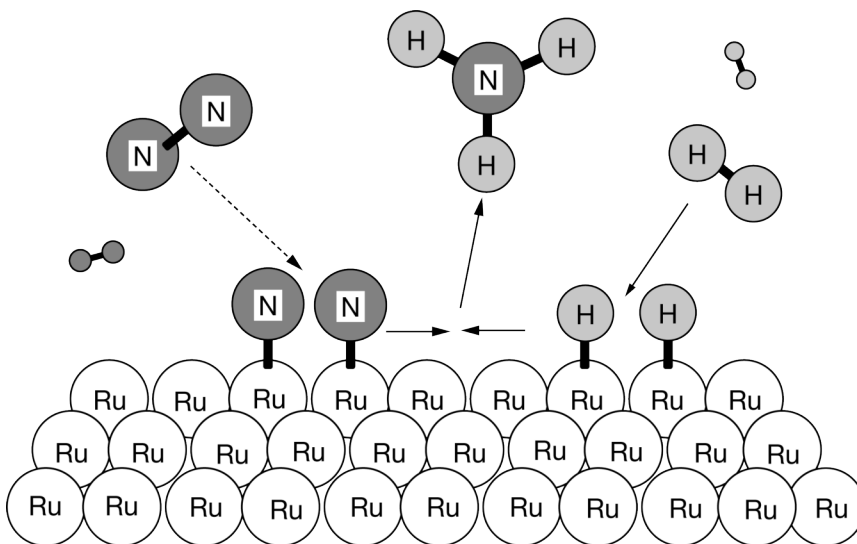
Time (s)	Volume of $\text{O}_2(g)$ Collected (mL)
0	0
60	42
120	64
180	76
240	82
300	86
360	88
420	89

49. The student observes that under the same conditions, the reaction occurs at a much slower rate when $\text{MnO}_2(s)$ is not used. Which of the following statements best explains this observation?

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- (A) $\text{MnO}_2(s)$ changes the reaction from endothermic to exothermic.
- (B) $\text{MnO}_2(s)$ shifts the equilibrium position of the reaction from reactant favored to product favored.
- (C) $\text{MnO}_2(s)$ releases heat in order to start the reaction.
- (D) $\text{MnO}_2(s)$ provides a reaction pathway with a lower activation energy. ✓

50.

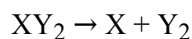


The diagram above illustrates how the reaction $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$ occurs on the surface of $\text{Ru}(s)$. The rate of this reaction is determined by the amount of energy required to break the bond in N_2 . This bond is weakened when N_2 is adsorbed on $\text{Ru}(s)$. Based on this information, which of the following provides the best reason for the use of $\text{Ru}(s)$ for the synthesis of NH_3 ?

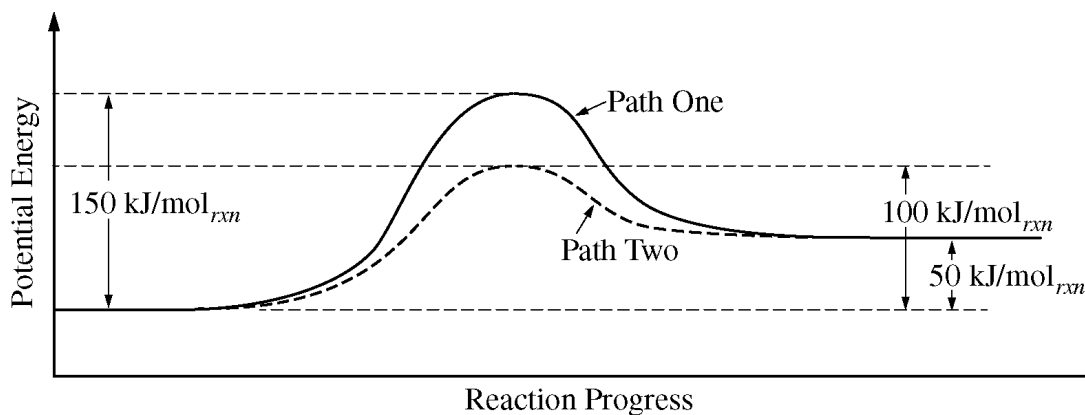
- (A) It promotes the proper orientation of N and H atoms to form new N – H bonds.
- (B) It provides a reaction path with a lower activation energy. ✓
- (C) It forms a bond between Ru and N that is stronger than the bond in N_2 .
- (D) It decreases the frequency of collisions between $\text{N}_2(g)$ and $\text{H}_2(g)$.

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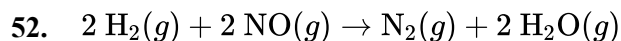
The following questions relate to the below information.



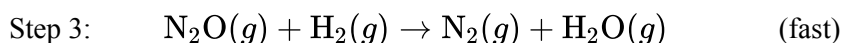
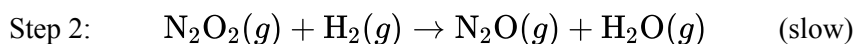
The equation above represents the decomposition of a compound XY_2 . The diagram below shows two reaction profiles (path one and path two) for the decomposition of XY_2 .



51. Which of the following most likely accounts for the difference between reaction path one and reaction path two?
- (A) A higher temperature in path one
 - (B) A higher temperature in path two
 - (C) The presence of a catalyst in path one
 - (D) The presence of a catalyst in path two



For the chemical reaction represented above, the following mechanism is proposed.



Which of the following initial rate law expressions is consistent with this proposed mechanism?

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(A) $\text{Rate} = k[\text{H}_2]^2[\text{NO}]^2$

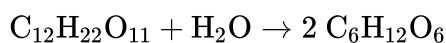
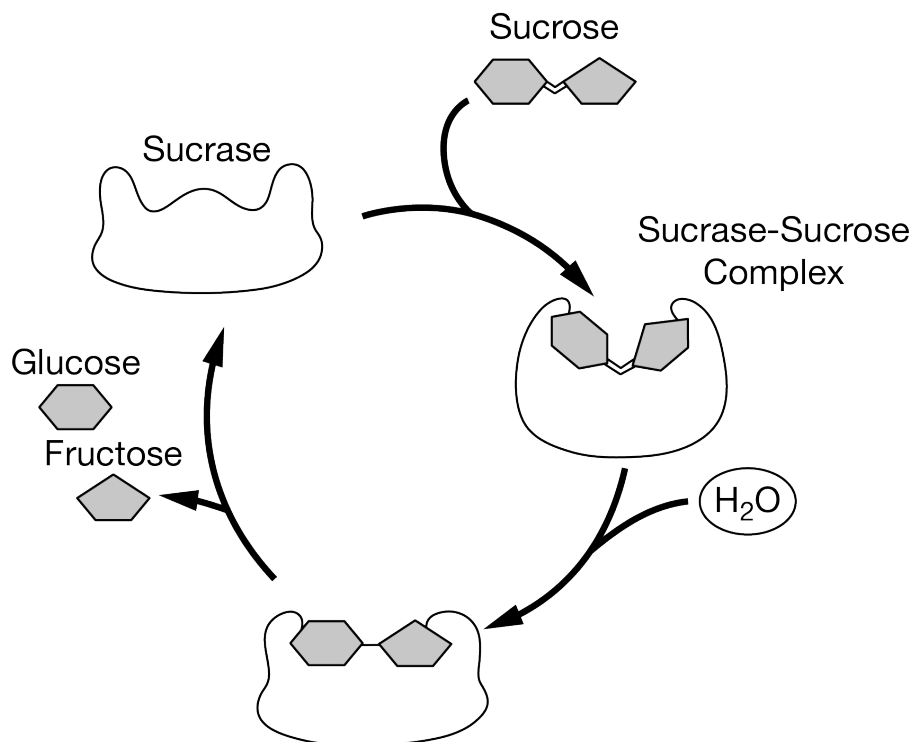
(B) $\text{Rate} = k[\text{NO}]^2$

(C) $\text{Rate} = k[\text{H}_2][\text{NO}]^2$

(D) $\text{Rate} = k[\text{H}_2][\text{N}_2\text{O}]^2$



53.



The hydrolysis of sucrose is represented by the chemical equation above. This reaction is extremely slow in aqueous solution. However, when sucrase is added as shown in the diagram, the rate of the reaction is about 6,000,000 times faster. Based on this information, which of the following best explains the large increase in the rate of hydrolysis that occurs with the addition of sucrase?

(A) The reaction proceeds through a different reaction path with a lower activation energy in which the sucrose-sucrased complex is formed as an intermediate.



(B) The addition of the sucrase reduces the number of effective collisions between the water and sucrose molecules.

(C) The reaction proceeds through a different reaction path that is independent of temperature and concentration.

(D) The addition of the sucrase decreases the number of sucrose molecules that have the minimum energy required to overcome the activation energy barrier.

54. The role of a catalyst in a chemical reaction is to

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- (A) decrease the amount of reactants that must be used
- (B) lower the activation energy for the reaction
- (C) supply the activation energy required for the reaction to proceed
- (D) increase the amounts of products formed at equilibrium
- (E) increase the entropy change for the reaction



Chapter 09 Part 3: Reaction Mechanism

55. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student investigates the physical and chemical properties of various carbon-containing compounds. The complete Lewis electron-dot diagrams and boiling points for two compounds, X and Z, are shown in the following table.

Compound	X	Z
Diagram	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & :\text{O}: & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{O}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & & & \end{array} $	$ \begin{array}{ccccccc} & \text{H} & \text{H} & \text{H} & \text{H} & & \\ & & & & & & \\ \text{H} & -\text{C} & -\text{C} & -\text{C} & -\text{C} & -\ddot{\text{S}} & -\text{H} \\ & & & & & & \\ & \text{H} & \text{H} & \text{H} & \text{H} & & \end{array} $
Boiling Point	437 K	371 K

- (a) Based on VSEPR theory, what is the approximate value of the O — C — O bond angle in compound X?
- b) The O — H bond in compound X is shorter than the S — H bond in compound Z. Explain the reason for this difference using principles of atomic structure.
- (c) For each compound, list all intermolecular forces present.
- Compound X —
Compound Z —
- (d) Explain why the boiling point of compound X is higher than that of compound Z based on the types and relative strengths of the intermolecular forces present in each substance.
- (e) A gaseous sample of compound Z is placed in an evacuated, rigid container at a temperature of 800. K and a pressure of 1.00 atm. Determine the final pressure of the gas when the temperature is decreased to 600. K.

Another carbon-containing compound is C₂H₆O, which reacts with H₂CrO₄, as represented by the following equation.



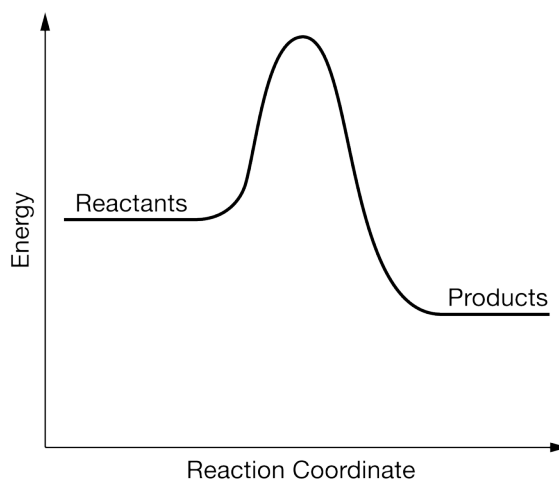
- (f) The student claims that C is reduced when C₂H₆O is converted to C₂H₄O. Do you agree or disagree? Justify your answer in terms of oxidation numbers.

The reaction is known to be exothermic. A proposed mechanism for the reaction is shown in the following table.

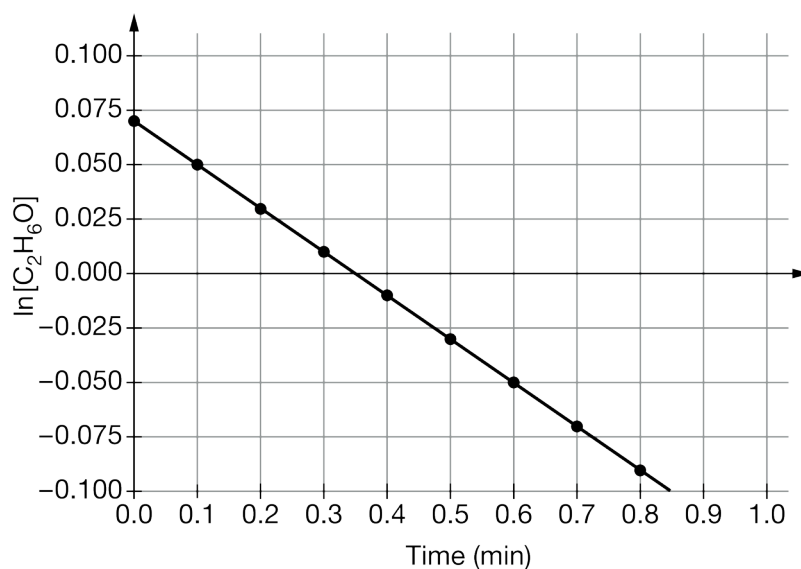
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Step 1	$\text{C}_2\text{H}_6\text{O} + \text{H}_2\text{CrO}_4 \rightleftharpoons \text{C}_2\text{H}_5\text{CrO}_4\text{H} + \text{H}_2\text{O}$	<i>fast</i>
Step 2	$\text{C}_2\text{H}_5\text{CrO}_4\text{H} \rightarrow \text{C}_2\text{H}_4\text{O} + \text{H}_2\text{CrO}_3$	<i>slow</i>

(g) The student draws the following energy diagram for the reaction. Is the drawing consistent with the proposed mechanism? Justify your answer.



(h) The reaction is run in an excess of H_2CrO_4 such that $[\text{H}_2\text{CrO}_4]$ remains constant. The student claims that under these conditions, the reaction is zeroth-order with respect to $\text{C}_2\text{H}_6\text{O}$, and that the rate constant of the reaction is 0.20 M/min . Is the following graph consistent with the student's claim? Justify your answer with respect to the graph's axes and the numerical value of the slope of the line.



Part A

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Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct hybridization.

- The bond angle is 120° .

Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a valid explanation equivalent to the following.

- The atomic radius of oxygen is smaller than that of sulfur because the valence electrons of oxygen occupy a lower shell than those of sulfur ($n = 2$ versus $n = 3$). Thus, the internuclear distance (bond length) is shorter for the $\text{O} - \text{H}$ bond than for the $\text{S} - \text{H}$ bond.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response correctly identifies the intermolecular forces in both compounds.

- Compound X – dispersion forces, dipole-dipole forces, and hydrogen bonding
- Compound Z – dispersion forces and dipole-dipole forces

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct explanation equivalent to the following.

· The dispersion forces in each compound should be similar because they have a comparable number of electrons and similarly polarizable electron cloud. However, compound X has relatively stronger dipole-dipole forces (due to the greater number of atoms with high electronegativity) and also hydrogen bonding. Because the total of all intermolecular forces in compound X is greater than that in compound Z, compound X has the higher boiling point.

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct calculation.

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{1.00 \text{ atm}}{800. \text{ K}} = \frac{P_2}{600. \text{ K}}$$

$$P_2 = 0.750 \text{ atm}$$

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

· Disagree. The oxidation number of carbon changes from -2 to -1 , indicating that it is oxidized, not reduced.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides correct answer and a valid explanation equivalent to the following.

· No. The lower energy of the products relative to the reactants is consistent with an exothermic reaction, but the diagram contains too few peaks. The energy diagram of a mechanism with two elementary steps would contain two peaks, not one.

Chapter 09 Part 3: Reaction Mechanism**Part H**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

· No. The linear relationship between $\ln [\text{C}_2\text{H}_6\text{O}]$ versus time is consistent with a first-order reaction, not a zeroth-order reaction.

OR

· No. The slope of the line is -0.20 min^{-1} , which is numerically consistent with the student's claim about the rate constant but inconsistent with the units of a zeroth-order reaction. In a zeroth-order reaction, $\text{slope} = -k$, so the units of the rate constant must be M/min rather than just min^{-1} .